

Linear Algebra 2H

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- *Last compiled: January 2026*
- Blended from notes of C. Keaton and J. R. Parker, Durham
- This was part of the Durham core second year modules. Involves introduction to vector spaces, linear maps, and matrices as linear maps.
- Course should really be “Vector spaces and linear algebra”. I have re-ordered things quite a bit from the original notes. *ordering to be tidied up...*

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1 *Vector spaces and linear maps*

Need a few definitions first; these are elaborated on and revisited later.

1.1 *Vector spaces*

A set V is a **vector space** over a field $\mathbb{F}$ if, for $v, u \in V$ and $\lambda, \mu \in \mathbb{F}$,

- V form an abelian group under addition $+$
- for all $\lambda v \in V$
- $(\lambda + \mu)v = \lambda v + \mu v$
- $\lambda(u + v) = \lambda u + \lambda v$
- $\lambda(\mu v) = (\lambda\mu)v$
- there exists an identity element 1 where $1v = v$.

Some examples are

- $\mathbb{F} = \mathbb{R}$ and $V = \mathbb{C}$
- $\mathbb{F} = \mathbb{R}$ and $V = \mathbb{R}^n$ is the Euclidean n -space
- $\mathbb{F} = \mathbb{R}$ and $V = C[0, 1]$ the set of continuous functions defined over the interval $[0, 1]$
- $\mathbb{F} = \mathbb{R}$ and V the solutions to the ODE $y'' - y' + y = 0$
- $\mathbb{F} = \mathbb{R}$ and $V = \mathbb{R}[t]$, the set of polynomials in t with coefficients in $\mathbb{R}$

Lemma 1.1.1 *If V is a vector space over $\mathbb{F}$, then for all $v \in V$, we have $0v = \mathbf{0}$.*

Proof We have $0v = (0 + 0)v = 0v + 0v$, so subtracting accordingly leads to the above result. ■

Lemma 1.1.2 *As above, but for all $\lambda \in \mathbb{F}$, we have $\lambda\mathbf{0} = \mathbf{0}$.* □

Lemma 1.1.3 For all $v \in V$, $(-1)v = -v$, the additive inverse of v .

Proof $0 = 0v = (1 + (-1))v = v + (-1)v$, so that $(-1)v = -v$. ■

Let $U \subseteq V$, then U is a **subspace** of V over $\mathbb{F}$ if U forms a vector space under the same addition and scalar multiplication as V .

Instead of checking that a given subspace U really satisfies all the vector space axioms, the follow theorem reduces the amount of work required.

Theorem 1.1.4 U is a subspace of V if and only if $U \neq \emptyset$, and for all $u, v \in U$ and $\lambda \in \mathbb{F}$, we have $u + v \in U$ and $\lambda v \in U$.

Proof These conditions are obviously necessary for U to be a vector space. Conversely, if $u, v \in U$, then $u, -v = (-1)v \in U$ by above lemma.s Since $u - v \in U$, $U \neq \emptyset$, and U is a group under addition. The condition $\lambda v \in U$ ensures that U is closed. Since the relevant axioms hold in V , they also hold in U . ■

Corollary 1.1.5 Let U and V be subspaces of W , then $U \cap V$ is a subspace of W .

1.2 Bases

We say the vectors $\{v_i\}$ are **linearly independent** if

$$\lambda_1 v_1 + \dots + \lambda_n v_n = 0 \quad (1.1)$$

only has trivial solutions $\lambda_i = 0$ for all i , if $v_i \neq v_j$ for $i \neq j$.

Example 1. Show

$$S = \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} \right\}$$

are linearly independent.

We have $a + 2c = 0$, $b = 0$, and $a + b + c = 0$. This ends up reducing down accordingly to $a = b = c = 0$, so S is linearly independent.

2. Shown

$$S = \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} \right\}$$

is linearly dependent.

Note that the third vector is the sum of the first and second vector.

3. Show that $S = \{\sin nx \mid n \in \mathbb{N}\}$ is linearly independent.

Note that we have the following orthogonality rule:

$$\int_{-\pi}^{\pi} \sin px \sin qx \, dx = \begin{cases} \pi, & p = q, \\ 0, & p \neq q. \end{cases}$$

Essentially from compound angle formula.

Suppose there exists some $\{b_i\}$ where $\sum_{k=1}^N b_k \sin kx = 0$ then we have, for some $p \in \{1, \dots, N\}$,

$$\sum_{k=1}^N b_k \int_{-\pi}^{\pi} \sin px \sin kx \, dx = 0,$$

so by the orthogonality condition, then $b_p \pi = 0$, and hence $b_p = 0$ for arbitrary p , and so we have linear independence.

This is one step towards saying sines and cosines forms the Fourier basis, which is important for the theorem of *Fourier transforms*.

The **span** of $\{v_i\}$ is defines as the set containing all linear combinations of the **spanning set** $\{v_i\}$, i.e.

$$\text{Span}(\{v_i\}) = \left\{ \sum_{i=1}^n \lambda_i v_i \mid \lambda_i \in \mathbb{F} \right\}. \quad (1.2)$$

Example 1. $S = \{e_1, e_2, (2, 3, 0)\}$ spans only the xy plane, because the third vector is not linearly independent of the other two.

2. Does $S = \{\sin nx \mid n \in \mathbb{N}\}$ span $C^\infty(\mathbb{R})$?

Suppose a function $f \in \text{Span}(S)$, then if S spans $C^\infty(\mathbb{R})$, we should have, for all $x \in \mathbb{R}$,

$$f(x) = \sum_{k=1}^n b_k \sin kx.$$

Here $f(0) = 0$, but note that $\cos(x) \in C^\infty(\mathbb{R})$, but $\cos 0 = 1$, so $g \notin \text{Span}(S)$, so the answer is no by a counter-example.

The set $\{v_i\}$ is a **basis** of some vector space U if the spanning set is linearly independent, and spans U . For

$$v = \sum_{i=1}^n v_i v_i, \quad (1.3)$$

we say v has **co-ordinates** $(v_1, \dots, v_n)$ with respect to the basis $\{v_i\}$.

Lemma 1.2.1 Let $S = \{v_1, \dots, v_n\}$ be a basis for a non-trivial V over $\mathbb{F}$. Then every non-zero element of V can be expressed uniquely by a linear combination of elements of S .

Proof Let $v \neq 0 \in V$. Since S is a spanning set, we have $v = \sum_{k=1}^n \lambda_k v_k$ for some constants $\lambda_i \in \mathbb{F}$. Suppose v has a different representation, i.e. $v = \sum_{k=1}^n \mu_k v_k$ for $\mu_k \in \mathbb{F}$, then we must have

$$\sum_{k=1}^n (\lambda_k - \mu_k) v_k = 0,$$

but since S is linearly independent, we must have $\lambda_k = \mu_k$, and so we have a unique representation. ■

Theorem 1.2.2 Let $A = \{u_1, \dots, u_m\}$ and $B = \{v_1, \dots, v_n\}$ be bases of V . Then $n = m$.

Proof The n or $m = 0$ case is trivial. Without loss of generality, assume $0 < m \leq n$. Since B is a basis for V , the first element of A could be represented as $u_1 = \sum_{k=1}^n \lambda_k v_k$. Since $u_1 \neq 0$, the λ are not all zero. Without loss of generality, suppose $\lambda_1 \neq 0$, then

$$v_1 = \frac{1}{\lambda_1}(u_1 - \lambda_2 v_2 - \dots - \lambda_n v_n),$$

so $v_1 \in \text{Span}(\{u_1, v_2, \dots, v_n\})$. The spanning set here is linearly independent, since if it wasn't, then there exists some μ_i where

$$\mu_1 u_1 + \mu_2 v_2 + \dots + \mu_n v_n = 0,$$

and following the argument above, we can assume $\mu_1 \neq 0$ without loss of generality since B is linearly independent, so $u_1 \in \text{Span}(\{v_2, \dots, v_n\})$, but by lemma, the representation of u_1 is unique, and since we assumed $\lambda_1 \neq 0$, we get a contradiction.

Note also that

$$\text{Span}(\{u_1, v_1, \dots, v_n\}) = \text{Span}(\{u_1, v_2, \dots, v_n\}) = V,$$

so $\{u_1, v_2, \dots, v_n\}$ serves as a basis for V . We can then continue the argument above for $v_2, \dots, v_n$, and if $m < n$, we reach the conclusion that

$$C = \{u_1, \dots, u_m, v_{m+1}, \dots, v_n\}$$

is a basis for V . However, since A is also a basis of V , is $v_n \in \text{Span}(A)$, then C would not be linearly independent, and we have a contradiction. So $m < n$ is false, and $m = n$ is the only possibility. ■

If a basis for a vector space V has n elements, then the vector space **dimension** $\dim(V) = n$.

Corollary 1.2.3 Let V be a n -dimensional vector space over $\mathbb{F}$, and $A = \{e_1, \dots, e_r\}$ be a subset of V which is linearly independent. Then A can be extended to a basis $\{e_1, \dots, e_n\}$ of V , where $r \leq n$.

Proof Let $B = \{v_1, \dots, v_n\}$ be a basis of V . If $\text{Span}(A) = V$ then we are done, otherwise there exists $e_{r+1} \in B$ such that $e_{r+1} \notin \text{Span}(A)$.

Suppose $\lambda_1 e_1 + \dots + \lambda_{r+1} e_{r+1} = 0$ for some $\lambda_i \in \mathbb{F}$. If $\lambda_{r+1} \neq 0$ then $e_{r+1} \in \text{Span}(A)$, which is a contradiction, so $\lambda_{r+1} = 0$, hence $A \cup \{e_{r+1}\}$ is a linearly independent set. We continue as necessary until we get $\{e_1, \dots, e_n\}$. ■

Corollary 1.2.4 Let V be a vector space over $\mathbb{F}$ with finite dimensions, and U a subspace of V . Then $\dim(U) \leq \dim(V)$, and if $\dim(U) = \dim(V)$, then $U = V$.

Proof Assume U is non-trivial, so there exists non-zero $e_1 \in U$. If $\text{Span}(e_1) = U$ then we are done, but otherwise there exists $e_2 \in U - \text{Span}(e_1)$, and by previous theorem, $\{e_1, e_2\}$ is a linearly independent set. We can continue as necessary for $r \leq n$, and by previous corollary we can continue until a basis of U is obtained. If $\dim(U) = \dim(V)$ then $r = n$, so $U = V$. ■

Corollary 1.2.5 Let $A = \{e_1, \dots, e_r\}$ be a subset of vector space V over $\mathbb{F}$. There exists a subset which is a basis of $\text{Span}(A)$, and hence $\dim(\text{Span}(A)) \leq r$.

Proof If A is linearly independent it is a basis, so $\dim(\text{Span}(A)) = r$. Otherwise, there exists non-trivial $\lambda_i \in \mathbb{F}$ such that $\sum_{i=1}^n \lambda_i e_i = \mathbf{0}$. Assume $\lambda_r \neq 0$, then $e_r \in \text{Span}(A - \{e_r\})$. If $\text{Span}(A - \{e_r\})$ is linearly independent then it is a basis and $\dim(\text{Span}(A - \{e_r\})) = r - 1 < r$ as. Otherwise continue as necessary. ■

Example 1. Find a subset of

$$S = \left\{ \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \right\}$$

which is a basis of $\text{Span}(S)$.

Notice that the second vector is a linear combination of the other two, so we could discard that to get S' which serves as a basis for $\text{Span}(S)$.

2. Extend S' to make it a basis for $\mathbb{R}^3$.

Note that for e_i , only e_1 is not in $\text{Span}(S')$, so we can augment S' with e_1 .

1.3 Linear maps

A **linear map** $\theta : U \rightarrow V$ for vector spaces over $\mathbb{F}$ is a function that satisfies

1. $\theta(u + v) = \theta(u) + \theta(v)$ for all $u, v \in U$,
2. $\theta(\lambda u) = \lambda \theta(u)$ for all $u \in U, \lambda \in \mathbb{F}$.

Note that θ is a *homomorphism*, respecting scalar multiplication.

Lemma 1.3.1

The composition of linear maps is linear.

Proof We have

$$\begin{aligned}\phi \circ \theta(\mathbf{u} + \mathbf{v}) &= \phi(\theta(\mathbf{u} + \mathbf{v})) = \phi(\theta(\mathbf{u}) + \theta(\mathbf{v})) \\ &= \phi(\theta(\mathbf{u})) + \phi(\theta(\mathbf{v})) \\ &= \phi \circ \theta(\mathbf{u}) + \phi \circ \theta(\mathbf{v}),\end{aligned}$$

and

$$\phi \circ \theta(\lambda \mathbf{u}) = \phi(\theta(\lambda \mathbf{u})) = \phi(\lambda \theta(\mathbf{u})) = \lambda \phi(\theta(\mathbf{u})) = \lambda \phi \circ \theta(\mathbf{u}).$$

■

Suppose $\theta : U \rightarrow V$ is a linear map, then we define

- **kernel** of θ as

$$\text{Ker}(\theta) = \{\mathbf{u} \in U \mid \theta(\mathbf{u}) = \mathbf{0}\}, \quad (1.4)$$

- **image** of θ as

$$\text{Im}(\theta) = \{\theta(\mathbf{u}) \mid \theta(\mathbf{u}) \in U\}. \quad (1.5)$$

Kernel is the set things that get sent to zero, while the image is the set of things that gets mapped by θ .

Lemma 1.3.2 The kernel and image of θ are subspaces of U and V respectively.

Proof Suppose $\mathbf{u}, \mathbf{v} \in \text{Ker}(\theta)$, then note that

$$\mathbf{0} = \theta(\mathbf{u}) - \theta(\mathbf{v}) = \theta(\mathbf{u} - \mathbf{v})$$

since θ is a homomorphism, so that $\mathbf{u} - \mathbf{v} \in \text{Ker}(\theta)$, while

$$\mathbf{0} = \lambda \mathbf{0} = \lambda \theta(\mathbf{u}) = \theta(\lambda \mathbf{u})$$

by property of a linear map, so $\lambda \mathbf{u} \in \text{Ker}(\theta)$. So we have closure and therefore the kernel is a subspace of U .

Suppose $\mathbf{a}, \mathbf{b} \in \text{Im}(\theta)$, then there exists $\mathbf{p}, \mathbf{q} \in U$ where $\theta(\mathbf{p}) = \mathbf{a}$ and $\theta(\mathbf{q}) = \mathbf{b}$. Then similar manipulations shows the image is closed, and is therefore a subspace of V . ■

Let $\theta : U \rightarrow V$ be a linear map.

- θ is a **monomorphism** if θ is injective, i.e. $\theta(\mathbf{u}) = \theta(\mathbf{v})$ implies that $\mathbf{u} = \mathbf{v}$.
- θ is an **epimorphism** if θ is surjective, i.e. $\text{Im}(\theta) = V$.
- θ is an **isomorphism** if θ is a bijection.
- θ is an **endomorphism** if $U = V$.

- θ is an **automorphism** if $U = V$ and θ is an isomorphism, i.e. it maps onto itself in a bijective way.

Lemma 1.3.3 *Let $\theta : U \rightarrow V$ be a linear map, then θ is a monomorphism if $\ker\theta = \{0\}$. θ is an isomorphism if $\text{Im}(\theta) = V$ and $\text{Ker}(\theta) = \{0\}$.*

Proof For the first part, if $u \in \text{Ker}(\theta)$, then $\theta(u) = \theta(0)$, and since θ is injective, we have $u = 0$. For the converse, if $\text{Ker}(\theta) = \{0\}$, then $\theta(u) = \theta(v)$ implies that $\theta(u - v) = 0$, so $u - v \in \text{Ker}(\theta)$. But the kernel is just the zero element, so this implies $u = v$, so we have a monomorphism. The isomorphism part follows directly from the first part by definition. ■

Let $\theta : U \rightarrow V$ be a linear map and U is finite dimensional. The **nullity** and **rank** of θ is the dimension of the kernel and image, i.e.

$$n(\theta) = \dim(\text{Ker}(\theta)), \quad r(\theta) = \dim(\text{Im}(\theta)). \quad (1.6)$$

Corollary 1.3.4 *θ is a monomorphism if $n(\theta) = 0$, and θ is an epimorphism iff $r(\theta) = \dim(V)$.* □

Theorem 1.3.5 *Let $\theta : U \rightarrow V$ be a linear map, and U is finite dimensional. Then $\dim(U) = r(\theta) + n(\theta)$.*

Proof Let $\{e_1, \dots, e_n\}$ be a basis for $\text{Ker}(\theta)$ where $n = n(\theta)$. This can be extended to another basis $\{e_1, \dots, e_n, w_1, \dots, w_r\}$ of U , and we aim to show that $\{w_i\}$ is a basis and $r = r(\theta)$.

The image of this basis under θ is then $\{0, \theta(w_1), \dots, \theta(w_r)\}$, so $\{\theta(w_1), \dots, \theta(w_r)\}$ spans $\theta(U)$. Suppose then $\lambda_1\theta(w_1) + \dots + \lambda_r\theta(w_r) = 0$. Since θ is linear, $\theta(\lambda_1w_1 + \dots + \lambda_rw_r) = 0$ and is in the kernel, and thus a change of basis implies that we should have some $\mu_i \in \mathbb{F}$ where

$$\lambda_1w_1 + \dots + \lambda_rw_r = \mu_1e_1 + \dots + \mu_ne_n,$$

and their difference is zero. The collection of w_i and e_i is a basis, and thus λ_i and μ_i are zero. Thus $\{w_i\}$ is linearly independent and thus a basis with $r = r(\theta)$, and we have $r + n = \dim(U)$. ■

Lemma 1.3.6 *Let $\{e_1, \dots, e_n\}$ be a basis of U and $\{f_1, \dots, f_n\}$ be a basis of V . Then there exists a unique linear map $\theta : U \rightarrow V$ such that $\theta(e_i) = f_i$ for each index.*

Proof For all $u \in U$, we have by definition $u = \lambda_i e_i$. Suppose $\theta(u) = \lambda_i f_i$, and if $v = \mu_i e_i$, then

$$\theta(u + v) = \theta((\lambda_i + \mu_i)e_i) = (\lambda_i + \mu_i)f_i = \lambda_i f_i + \mu_i f_i = \theta(u) + \theta(v),$$

and that

$$\theta(vu) = \theta(v(\lambda_i e_i)) = v\theta(u),$$

hence θ is a linear map.

Suppose there exists from other linear map ϕ satisfying the conditions. Then

$$\theta(\mathbf{u}) = \lambda_i \theta(\mathbf{e}_i) = \lambda_i \phi(\mathbf{e}_i) = \phi(\lambda_i \mathbf{e}_i) = \phi(\mathbf{u}),$$

so we have uniqueness as required. $\blacksquare$

Example For $f : \mathbb{R}^3 \rightarrow \mathbb{R}^3$, let

$$f(\mathbf{e}_1) = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \quad f(\mathbf{e}_2) = \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix}, \quad f(\mathbf{e}_3) = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}.$$

Compute the nullity and rank of f .

Note that $f(x, y, z) = xf(\mathbf{e}_1) + yf(\mathbf{e}_2) + zf(\mathbf{e}_3)$ by definition, so the set of simultaneous equations we have are

$$-x + 3y + 2z = v_1, \quad x - y + z = v_2, \quad y + z = v_3.$$

If $\mathbf{v} = \mathbf{0}$, then $y = -z$, which implies $x = -2z$, which implies $z = y = x = 0$, so the map here has a trivial kernel and is full rank by Thm. 1.3.5, i.e. $n(f) = 0$ and $r(f) = 3$.

1.4 Sums and direct sums

Let U and V be vector spaces over $\mathbb{F}$. Then **sum** of U and V is

$$U + V = \{\mathbf{u}, \mathbf{v} \mid \mathbf{u} \in U, \mathbf{v} \in V\}. \quad (1.7)$$

Lemma 1.4.1 *If U and V are subspaces of W , then $U + V$ is also a subspace of W .*

Proof $U + V$ is non-empty since $\mathbf{0} + \mathbf{0} = \mathbf{0} \in U + V$. Let $\mathbf{u}, \mathbf{u}' \in U$ and $\mathbf{v}, \mathbf{v}' \in V$, then

$$\lambda(\mathbf{u} + \mathbf{v}) = \lambda\mathbf{u} + \lambda\mathbf{v} \in U + V$$

for all $\lambda \in \mathbb{F}$. Also,

$$(\mathbf{u} + \mathbf{v}) + (\mathbf{u}' + \mathbf{v}') = (\mathbf{u} + \mathbf{u}') + (\mathbf{v} + \mathbf{v}') \in U + V,$$

so $U + V$ is also closed, and thus a subspace.

The following theorem is useful for determining dimensions of spaces.

Theorem 1.4.2 *Let U and V be finite subspaces of W over $\mathbb{F}$. Then*

$$\dim(U) + \dim(V) = \dim(U \cap V) + \dim(U + V). \quad (1.8)$$

Proof By Corollary 1.2.3, there exists a basis $B = \{e_1, \dots, e_r\}$ which is a basis of $U \cap V$ that may be extended to $\{e_1, \dots, e_r, f_1, \dots, f_s\}$ of U , and $\{e_1, \dots, e_r, g_1, \dots, g_t\}$ of V .

Clearly $\{e_1, \dots, e_r, f_1, \dots, f_s, g_1, \dots, g_t\}$ spans $U + V$. Suppose there exists non-trivial scalars λ_i, μ_i and ν_i such that

$$\sum_{i=1}^r \lambda_i e_i + \sum_{i=1}^s \mu_i f_i + \sum_{i=1}^t \nu_i g_i = \mathbf{0}.$$

Then not all of ν_i is zero, since $\{e_1, \dots, e_r, f_1, \dots, f_s\}$ is a basis of U and hence linearly independent (and similarly for the other case).

Then $\{g_1, \dots, g_t\}$ is linearly independent and hence in U , so much be in $U \cap V$. Then

$$\sum_{i=1}^t \nu_i g_i = \sum_{i=1}^r \lambda_i e_i,$$

but this cannot be since $\{e_1, \dots, e_r, g_1, \dots, g_t\}$ is a basis of V . So then $\{e_1, \dots, e_r, f_1, \dots, f_s, g_1, \dots, g_t\}$ is a basis of $U + V$. Counting dimensions accordingly leads to the result. ■

Example Find a basis of $U \cap V$, and hence determine $\dim(U)$, $\dim(V)$, $\dim(U \cap V)$, and $\dim(U + V)$, where

$$U = \text{Span} \left(\left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix} \right\} \right), \quad V = \text{Span} \left(\left\{ \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix} \right\} \right).$$

We see the spanning sets of U and V are linearly independent, hence $\dim(U) = \dim(V) = 2$. We note that $w \in U \cap V$ can be expressed as

$$w = a \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + b \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix} = c \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix} + d \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix},$$

and the resulting system implies that $a = c$ and $b = d = -c$. Taking $c = 1$ leads to

$$U \cap V = \text{span} \left(\begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix} \right),$$

and hence $\dim(U \cap V) = 1$. By Theorem 1.4.2, we should have $\dim(U + V) = 2 + 2 - 1 = 3$, which can also be checked directly.

The **direct sum** of U and V is denoted $U \oplus V$, which is $U + V$ but where $U \cap V = \{\mathbf{0}\}$.

Theorem 1.4.3 Let U and V be subspaces of a vector space. Then the $U + V$ is a direct sum if and only if there exists unique $u \in U$, $v \in V$ where for all $w = u + v \in W$.

Proof If $W = U \oplus V$ then $w = u + v$ by definition, so we need to shown uniqueness. Suppose $w = u' + v'$, then

$$u' + v' = u + v \quad \Leftrightarrow \quad v' - v = u - u'.$$

We note that $v' - v, u - u' \in U \cap V$, so by definition they are just the zero vector, and we have uniqueness.

Let $w \in U \cap V$ and assume we have uniqueness. So we must have $w = w + 0_V = 0_U + w$ for the respective zero vectors, but the above is only true if $w = 0$, and hence $U \cap V = \{0\}$ and we have a direct sum. ■

Example If $U = \text{Span}(\{e_1\})$ and $V = \text{Span}(\{e_3\})$ then $W = U \oplus V = \text{Span}(\{e_1, e_3\})$.

On the other hand, $U = \text{Span}(\{e_1 + e_2\})$ and $V = \text{Span}(\{e_2 + e_3\})$, the sum is not direct, and we have $W = U + V = \text{Span}(\{e_1, e_2, e_3\})$.

Let U and V be vector spaces over $\mathbb{F}$. Then $U \times V$ is the **external direct sum**, which is a vector space with the definition that the ordered pairs satisfy

$$(u, v) + (u', v') = (u + u', v + v'), \quad \lambda(u, v) = (\lambda u, \lambda v).$$

1.5 Inner products

An **inner product** on a real vector space V is a function

$$\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R} \tag{1.9}$$

such that we have the following properties:

- symmetry, $\langle u, v \rangle = \langle v, u \rangle$,
- linearity in the first factor, $\langle \lambda u + \mu w, v \rangle = \lambda \langle u, v \rangle + \mu \langle w, v \rangle$,
- positivity, $\langle v, v \rangle \geq 0$ and $\langle v, v \rangle = 0$ iff $v = 0$.

The inner product provides a notion of distance and angles between vector in vector spaces.

For the above, if we are instead dealing with a complex vector space then the first condition becomes

$$\langle u, v \rangle = \overline{\langle v, u \rangle}$$

where the overbar denotes a complex conjugate. Note then we also have

$$\langle u, \lambda v \rangle = \overline{\lambda} \langle u, u \rangle.$$

A function that is linear in the first factor and anti-linear like above in the second factor is called **sesquilinear**.

Example The standard scalar (dot) product in $\mathbb{R}^n$ is also known as the **Euclidean inner product**.

Example If we take $V = \mathbb{R}^n$ and $a_k \in \mathbb{R}^+$, then we can check that $\langle \mathbf{u}, \mathbf{v} \rangle = \sum_{i=1}^n a_i u_i v_i$ is an inner product.

Example For $V = \mathbb{R}^2$ and $a, b, c \in \mathbb{R}$, we consider symmetric matrices of the form

$$A = \begin{pmatrix} a & b \\ b & c \end{pmatrix}.$$

Find the conditions on a, b, c such that

$$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{v}^T A \mathbf{u} = a u_1 v_1 + b(u_1 v_2 + u_2 v_1) + c u_2 v_2$$

is an inner product.

Picking some useful vectors, if $\mathbf{u} = \mathbf{v} = \mathbf{e}_1$ then $\langle \mathbf{u}, \mathbf{u} \rangle = a$, so $a \geq 0$. Similarly, $\mathbf{u} = \mathbf{v} = \mathbf{e}_2$ implies $c \geq 0$. Now

$$\begin{aligned} \langle \mathbf{v}, \mathbf{v} \rangle &= a \left[v_1^2 + \frac{2b}{a} v_1 v_2 + \frac{c}{a} v_2^2 \right] \\ &= a \left[\left(v_1 + \frac{b}{a} v_2 \right)^2 - \frac{b^2}{a^2} v_2^2 + \frac{c}{a} v_2^2 \right] \\ &= a \left[\left(v_1 + \frac{b}{a} v_2 \right)^2 + \frac{ac - b^2}{a^2} v_2^2 \right] \geq 0 \end{aligned}$$

if $\mathbf{v} \neq \mathbf{0}$, so we additionally want $ac - b^2 = |A| > 0$.

Example If we take $V = C([a, b])$, show that

$$\langle f, g \rangle = \int_a^b f(t)g(t) \, dt$$

defines an inner product.

The linearity and symmetry properties clearly hold. $\langle f, f \rangle \geq 0$ is also true since the integrand $f^2(t)$ is a positive real valued function, and the integral is zero only if $f \equiv 0$ for all $t \in [a, b]$.

Example For $V = \mathbb{C}^n$, the **standard Hermitian inner product** is defined to be (with an implied sum)

$$\langle \mathbf{z}, \mathbf{w} \rangle = \mathbf{w}^\dagger \mathbf{z} = z_i \overline{w}_i \quad (1.10)$$

where the dagger denotes the Hermitian transpose.

Example Show that

$$\langle \mathbf{z}, \mathbf{w} \rangle = \mathbf{w}^\dagger \begin{pmatrix} 3 & i \\ -i & 2 \end{pmatrix} \mathbf{z}$$

is a Hermitian inner product on $\mathbb{C}^2$.

Since we are dealing with a single matrix the symmetry and linearity is satisfied, so we need positivity. We have

$$\begin{aligned}\langle \mathbf{z}, \mathbf{z} \rangle &= 3z_1\bar{z}_1 + iz_1\bar{z}_2 - iz_2\bar{z}_1 + 2z_2\bar{z}_2 \\ &= z_1(3\bar{z}_1 + i\bar{z}_2) - iz_2\bar{z}_1 + 2z_2\bar{z}_2 \\ &= \frac{1}{3}(3\bar{z}_1 + i\bar{z}_2)(3z_1 - iz_2) - \frac{1}{3}z_2\bar{z}_2 + 2z_2\bar{z}_2 \\ &= \frac{1}{3}|3z_1 - iz_2|^2 + \frac{5}{3}|z_2|^2 \geq 0\end{aligned}$$

and is only zero when $\mathbf{z} = \mathbf{0}$.

There are presumably other ways to cook up some squares via factorisation and multiply by 1 tricks.

1.6 Length, distance and angles

The **norm** of a vector is defined to be

$$\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}. \quad (1.11)$$

The **distance** between two vectors is

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|. \quad (1.12)$$

A **unit vector** $\hat{\mathbf{v}} = \mathbf{v}/\|\mathbf{v}\|$ has unit norm.

Lemma 1.6.1

$$\operatorname{Re} \langle \mathbf{u}, \mathbf{v} \rangle = \frac{1}{2} (\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u}\|^2 - \|\mathbf{v}\|^2).$$

The proof is straight-forward algebraic manipulations. $\square$

Lemma 1.6.2 (Cauchy-Schwarz inequality) We have

$$|\langle \mathbf{u}, \mathbf{v} \rangle|^2 \leq \|\mathbf{u}\|^2 \|\mathbf{v}\|^2, \quad (1.13)$$

with equality iff one of the entries is zero or they are parallel to each other.

Proof The zero case is trivial. Otherwise, for $\lambda \in \mathbb{R}$,

$$0 \leq \|\lambda \mathbf{u} + \mathbf{v}\|^2 = \lambda^2 \|\mathbf{u}\|^2 + 2\lambda \langle \mathbf{u}, \mathbf{v} \rangle + \|\mathbf{v}\|^2.$$

Let $\lambda = -\langle \mathbf{u}, \mathbf{v} \rangle / \|\mathbf{u}\|^2$, then

$$\begin{aligned}0 &\geq \frac{\langle \mathbf{u}, \mathbf{v} \rangle^2}{\|\mathbf{u}\|^2} - \frac{2\langle \mathbf{u}, \mathbf{v} \rangle^2}{\|\mathbf{u}\|^2} + \|\mathbf{v}\|^2 \\ &= -\frac{\langle \mathbf{u}, \mathbf{v} \rangle^2}{\|\mathbf{u}\|^2} + \|\mathbf{v}\|^2\end{aligned}$$

and the inequality follows. With have equality iff

$$\mathbf{v} = -\lambda \mathbf{u} = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\|^2} \mathbf{u},$$

i.e. $\mathbf{v}$ is a scalar multiple of $\mathbf{u}$ and therefore are parallel to each other.

■

Corollary 1.6.3 (Triangle inequality) *If a real inner product space we have $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$, and if neither entries are zero, then we have equality iff $\mathbf{v} = \lambda \mathbf{u}$ for some scalar λ .*

Proof Note that

$$(\|\mathbf{u}\| + \|\mathbf{v}\|)^2 - \|\mathbf{u} + \mathbf{v}\|^2 = 2\|\mathbf{u}\|\|\mathbf{v}\| - 2\langle \mathbf{u}, \mathbf{v} \rangle \geq 0$$

by Cauchy-Schwarz, and so the LHS gives the result. ■

Let $\mathbf{u}$ and $\mathbf{v}$ be two non-zero vectors in an inner product space. The **angle** between $\mathbf{u}$ and $\mathbf{v}$ is the solution to

$$\cos \theta = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\|\|\mathbf{v}\|}, \quad \theta \in [0, \pi]. \quad (1.14)$$

Two vectors are **orthogonal** if $\theta = \pi/2$, i.e. $\langle \mathbf{u}, \mathbf{v} \rangle = 0$.

Lemma 1.6.4 *If $\mathbf{u}$ and $\mathbf{v}$ are two orthogonal vectors then $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$. Follows from Lemma 1.6.1.* ■

This is basically Pythagoras' theorem but in a more general sense.

Example For $V = C([-1, 1])$, we have

$$\langle 1, x \rangle = \int_{-1}^1 1 \cdot x \, dx = 0$$

since the integrand is odd, so the identity function (which is non-zero) and x are orthogonal.

If we instead take $V = C([-1, 1])$, then we note that $\|1\| = 1$, $\|x\| = 1/\sqrt{3}$, and $\langle 1, x \rangle = 1/2$, and instead $\theta = \arccos(\sqrt{3}/2) = \pi/6$.

1.7 Gram-Schmidt orthonormalisation

Let V be an inner product space with a basis $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$. This basis is **orthonormal** if $\langle \mathbf{v}_i, \mathbf{v}_j \rangle = \delta_{ij}$.

Note that every such set of basis has an isomorphism to the standard basis in $\mathbb{R}^n$ or $\mathbb{C}^n$.

The **Gram-Schmidt orthonormalisation** is a systematic way of turning a basis into an orthonormal basis. The idea is to normalise one of these basis vector first to give the first entry, then project that component out of the second vector and normalise and get the second entry, and continue a finite number of times. It is not necessarily fast, but it will give what is needed. In symbols, what we do is the following:

$$\begin{aligned} \mathbf{u}_1 &= \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} \\ \mathbf{u}_2 &= \frac{\tilde{\mathbf{v}}_2}{\|\tilde{\mathbf{v}}_2\|}, \quad \tilde{\mathbf{v}}_2 = \mathbf{v}_2 - \langle \mathbf{v}_2, \mathbf{u}_1 \rangle \mathbf{u}_1 \\ \mathbf{u}_k &= \frac{\tilde{\mathbf{v}}_k}{\|\tilde{\mathbf{v}}_k\|}, \quad \tilde{\mathbf{v}}_k = \mathbf{v}_k - \langle \mathbf{v}_k, \mathbf{u}_{k-1} \rangle \mathbf{u}_{k-1} - \dots - \langle \mathbf{v}_k, \mathbf{u}_1 \rangle \mathbf{u}_1. \end{aligned}$$

Then $\{\mathbf{u}_1, \dots, \mathbf{u}_n\}$ is an orthonormal basis.

Gram-Schmidt is related to the **QR decomposition**, which is an important way to compute eigenvalues for example. In deployment however Gram-Schmidt is not numerically stable, and other variations are used instead.

Example Taking the standard Euclidean inner product in $\mathbb{R}^3$ and

$$v_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad v_3 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix},$$

we have

$$u_1 = \frac{v_1}{\|v_1\|} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}.$$

Then the projection step gives

$$\tilde{v}_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} - \frac{1}{\sqrt{2}} \left\langle \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \right\rangle \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix},$$

and normalising gives

$$u_2 = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}.$$

The third one can be shown to be

$$u_3 = \frac{1}{\sqrt{3}} \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}.$$

Example For the norm defined by the symmetric matrix

$$A = \begin{pmatrix} 4 & 1 \\ 1 & 1 \end{pmatrix},$$

note that $v_1 = e_1$ and $v_2 = e_2$ we would have $\|v_1\|^2 = 4 + 0 + 0 + 0$, so

$$u_1 = \frac{1}{2} \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

The projection step then gives

$$\tilde{v}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} - \left[\frac{1}{2} \begin{pmatrix} 0 & 1 \end{pmatrix} \begin{pmatrix} 4 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right] \frac{1}{2} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1/4 \\ 1 \end{pmatrix},$$

so that

$$u_2 = \frac{1}{\sqrt{1/4 - 1/4 - 1/4 + 1}} \begin{pmatrix} -1/4 \\ 1 \end{pmatrix} = \frac{2}{\sqrt{3}} \begin{pmatrix} -1/4 \\ 1 \end{pmatrix}.$$

Note that we know that there is a linear isomorphism $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ where $T(e_i) = u_i$. For the above example T can be represented by

$$B = \begin{pmatrix} 1/2 & -1/(2\sqrt{3}) \\ 0 & 2/\sqrt{3} \end{pmatrix}, \quad B^{-1} = \begin{pmatrix} 2 & 1/2 \\ 0 & \sqrt{3}/2 \end{pmatrix},$$

where B is formed by the orthonormal basis $\{u_1, u_2\}$. Then notice that

$$T^{-1}(v_1) = B^{-1}v_1 = \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \quad T^{-1}(v_2) = B^{-1}v_2 = \begin{pmatrix} 1/2 \\ \sqrt{3}/2 \end{pmatrix},$$

so that in the **standard** Euclidean norm (because we are in the original space where the standard basis live)

$$\cos \phi = \frac{\langle T^{-1}(v_1), T^{-1}(v_2) \rangle}{\|T^{-1}(v_1)\| \|T^{-1}(v_2)\|} = \frac{2 \cdot 1/2 + 0}{2 \cdot (1/4 + 3/4)} = \frac{1}{2},$$

which implies $\phi = \pi/3$. Also note that, in transformed space defined by the norm A ,

$$\cos \theta = \frac{\langle v_1, v_2 \rangle_A}{\|v_1\| \|v_2\|} = \frac{4 \cdot 1 \cdot 0 + 1 \cdot 1 \cdot 1 + 1 \cdot 1 \cdot 0 + 1 \cdot 0 \cdot 1}{\sqrt{4} \cdot \sqrt{1}} = \frac{1}{2},$$

and so $\theta = \pi/3 = \phi$ also. This is perhaps not a surprise if T is suppose to be an isomorphism (since it should then preserve angles), and is saying the result of Gram-Schmidt implies transformation; the resulting matrix would be *upper*-triangular, since the projection step removes the components so some entries (in this case below the diagonal) are zero.

Example We can do this for functions too. For $V = C([-1, 1])$, if we take the usual inner product and $v_1 = 1$, $v_2 = t$ and $v_3 = t^2$, then

$$u_1 = \frac{1}{\sqrt{2}},$$

$$\tilde{v}_2 = t - 0 \Rightarrow u_2 = \frac{2}{3}t$$

since 1 and t are orthogonal, and

$$\tilde{v}_3 = t^2 - 0 - \frac{\sqrt{2}}{3} \frac{1}{\sqrt{2}} \Rightarrow u_3 = \frac{5}{2\sqrt{2}}(3t^2 - 1)$$

since t^2 and t are orthogonal but t^2 and 1 are not.

Example Suppose we have

$$\langle z, w \rangle = 3z_1\bar{w}_1 + iz_1\bar{w}_2 - iz_2\bar{w}_1 + 2z_2\bar{w}_2,$$

Or is **conformal**. Note that isometry $\Rightarrow$ conformal but not vice-versa.

This also relates to the QR algorithm mentioned above.

These are basically the first three **Legendre polynomials** upto normalisation factors.

then $v_1 = e_1$ and $v_2 = e_2$ get orthonormalised to

$$u_1 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 0 \end{pmatrix},$$

$$\tilde{v}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} - \frac{1}{\sqrt{3}} (3 \cdot 1 \cdot 0 + i \cdot 0 \cdot 0 - i \cdot 1 \cdot 1 + 2 \cdot 0 \cdot 1) \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} i/3 \\ 1 \end{pmatrix},$$

so

$$u_2 = \frac{1}{3/9 + 2 - 1/3 - 1/3} \begin{pmatrix} i/3 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{5}} \begin{pmatrix} i \\ 3 \end{pmatrix}.$$

1.8 Orthogonal projection

Let U be a subspace of an inner product space V . The **orthogonal complement** is defined as

$$U^\perp = \{v \in V \mid \langle u, v \rangle = 0 \forall u \in U\}. \quad (1.15)$$

Lemma 1.8.1 Suppose we can write $v = u + \tilde{u}$ for $u \in U$ and $\tilde{u} \in U^\perp$. Then the decomposition is unique.

Proof Suppose $v = u + \tilde{u} = u' + \tilde{u}'$, then

$$0 = u - u' + \tilde{u} - \tilde{u}'$$

so that $u' - u = \tilde{u} - \tilde{u}' \in U^\perp$, but $u' - u \in U$, so $u' - u = 0$ and $u' = u$, which also implies $\tilde{u} = \tilde{u}'$. ■

Lemma 1.8.2 If U is finite dimensional then u and $\tilde{u}$ are guaranteed to exist.

Proof Let $\{u_1, \dots, u_k\}$ be an orthonormal basis of U . If we set $u = \langle u, u_i \rangle u_i$ (sum implied) and define $\tilde{u} = v - u$. Then $u \in U$ by construction, and $v = u + \tilde{u}$. We aim to show $\tilde{u} \in U^\perp$; it is enough to show $\langle \tilde{u}, u_i \rangle = 0$ for all i since we have a basis of U .

$$\begin{aligned} \langle \tilde{u}, u_i \rangle &= \langle v - u, u_i \rangle \\ &= \langle v, u_i \rangle - \langle \langle v, u_j \rangle u_j, u_i \rangle \\ &= \langle v, u_i \rangle - \langle v, u_j \rangle \langle u_j, u_i \rangle \\ &= \langle v, u_i \rangle - \langle v, u_j \rangle \delta_{ij} \\ &= \langle v, u_i \rangle - \langle v, u_i \rangle \\ &= 0, \end{aligned}$$

hence $\tilde{u} \in U^\perp$ ■

Proposition 1.8.3 Let U be a finite dimensional subspace of an inner product space V . Let $v_0 \in V$, then if $v_0 = u_0 + \tilde{u}_0$ for all $u \in U$, then $\|v_0 - u_0\| \leq \|v_0 - u\|$ with equality iff $u_0 = u$.

Proof

$$\begin{aligned}\|v_0 - u_0\|^2 &= \|v_0 - u_0 + u_0 - u\|^2 \\ &= \|v_0 - u_0\|^2 + \|u_0 - u\|^2 \\ &\geq \|v_0 - u_0\|\end{aligned}$$

with equality iff $u_0 - u = 0$, where the second line follows by Pythagoras' theorem since $u_0 - u \in U$ and $v_0 - u_0 \in U^\perp$. ■

u_0 is called the **orthogonal projection** of v_0 onto U .

Example Find the point in

$$\text{span} \left\{ v_1 = \begin{pmatrix} 0 \\ 2 \\ -1 \\ 2 \end{pmatrix}, v_2 = \begin{pmatrix} 2 \\ 3 \\ 1 \\ 2 \end{pmatrix} \right\}$$

closest to

$$v_0 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

with respect to the usual inner product in $\mathbb{R}^4$.

By Gram-Schmidt, we have

$$u_1 = \frac{1}{3} \begin{pmatrix} 0 \\ 2 \\ -1 \\ 2 \end{pmatrix}, \quad u_2 = \frac{1}{3} \begin{pmatrix} 2 \\ 1 \\ 2 \\ 0 \end{pmatrix}.$$

The orthogonal projection onto the above space is

$$u_0 = \langle v_0, u_1 \rangle u_1 + \langle v_0, u_2 \rangle u_2 = \frac{1}{3} \cdot \frac{1}{3} \begin{pmatrix} 0 \\ 2 \\ -1 \\ 2 \end{pmatrix} + \frac{1}{3} \cdot \frac{5}{3} \begin{pmatrix} 2 \\ 1 \\ 2 \\ 0 \end{pmatrix} = \frac{1}{9} \begin{pmatrix} 10 \\ 11 \\ 7 \\ 6 \end{pmatrix}.$$

The distance of v_0 from u_0 is $\|v_0 - u_0\|$, but since they are orthogonal then by Pythagoras we have

$$\|v_0 - u_0\|^2 = \|v_0\|^2 - \|u_0\|^2 = \|v_0\|^2 - \langle v_0, u_1 \rangle^2 - \langle v_0, u_2 \rangle^2$$

as u_1 and u_2 are basis vectors of U , so

$$\|v_0 - u_0\|^2 = 4 - 1 \frac{25}{9} = \frac{2}{9}.$$

Example For $V = C([-1, 1])$ with the usual integral as a inner product. Find the linear polynomial closest to $g(t) = t^2 + t$.

Let $U = \text{span}(\{1, t\})$. We know from a previous example that $\{1/\sqrt{2}, \sqrt{3/2}t\}$ is an orthonormal basis, so

$$\langle g, 1/\sqrt{2} \rangle = \frac{\sqrt{2}}{3}, \quad \langle g, \sqrt{3/2}t \rangle = \sqrt{\frac{2}{3}},$$

so the orthogonal project of g onto U is

$$f(t) = \frac{\sqrt{2}}{3} \frac{1}{\sqrt{2}} + \sqrt{\frac{2}{3}} \sqrt{\frac{3}{2}} t = \frac{1}{3} + t.$$

Lemma 1.8.4 (Bessel's inequality) Let V be a finite dimensional inner product space. If $\mathbf{v} \in V$ and $\mathbf{u} \in U$ where U is the orthogonal projection of $\mathbf{v}$ onto U , then $\|\mathbf{u}\|^2 \leq \|\mathbf{v}\|^2$. Furthermore, if $\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ is an orthonormal basis of U with $\mathbf{u} = \sum_{i=1}^k \lambda_i \mathbf{u}_i$, then $\sum_{i=1}^k \lambda_i^2 \leq \|\mathbf{v}\|^2$.

Proof Let $\mathbf{v} = \mathbf{u} + \tilde{\mathbf{u}}$, $\mathbf{u} \in U$ and $\tilde{\mathbf{u}} \in U^\perp$. By Pythagoras' theorem, $\|\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\tilde{\mathbf{u}}\|^2 \geq \|\mathbf{u}\|^2$. If $\mathbf{u} = \sum_{i=1}^k \lambda_i \mathbf{u}_i$, then again by Pythagoras

$$\|\mathbf{v}\|^2 = \|\sum_{i=1}^k \lambda_i \mathbf{u}_i\|^2 = \sum_{i=1}^k \lambda_i^2 \|\mathbf{u}_i\|^2 = \sum_{i=1}^k \lambda_i^2$$

since $\|\mathbf{u}_i\| = 1$.

2 Matrices

Mostly going to assume we are working with matrices defined over the reals, unless stated otherwise.

Let $\theta : U \rightarrow V$ be a linear map, and let $\{e_i\}$ and $\{f_i\}$ be a basis of U and V respectively. Then by Lemma 1.3.6, θ can be expressed in terms of a matrix

$$\theta(e_j) = a_{ij}f_i \quad (2.1)$$

where $A = (a_{ij})$.

Theorem 2.0.1 If $\theta : U \rightarrow V$ and $\phi : V \rightarrow W$ are linear maps represented by A and B respectively, with respect to the basis $\{e_i\}$, $\{f_i\}$ and $\{g_i\}$ of U , V and W , then $\phi \circ \theta$ is represented by BA with respect to the same bases.

Proof Let C represent $\phi \circ \theta : U \rightarrow W$, so that $(\phi \circ \theta)(e_k) = c_{ik}g_i$. But we also have

$$(\phi \circ \theta)(e_k) = \phi(\theta(e_k)) = \phi(a_{jk}f_j) = a_{jk}\phi(f_j) = a_{jk}b_{ij}g_i,$$

and equating the coefficients leads to $c_{ik} = a_{jk}b_{ij}$ with $C = BA$. ■

A linear map can be represented in different bases.

Theorem 2.0.2 Let $\theta : U \rightarrow V$ be represented by A with bases $\{e_1, \dots, e_n\}$ and $\{f_1, \dots, f_m\}$ respectively. If $\{e'_1, \dots, e'_n\}$ and $\{f'_1, \dots, f'_m\}$ are also bases of U and V , then θ is represented by XAY where

$$X : e'_i = y_{ki}e_k, \quad Y : f_i = x_{ki}f'_k.$$

Proof Note that

$$\theta(e_i) = \theta(y_{ki}e_k) = y_{ki}\theta(e_k) = y_{ki}a_{jk}f_j = y_{ki}a_{jk}x_{lj}f'_l = x_{lj}a_{jk}y_{ki}f'_l,$$

which is $(XAY)_{li}$. ■

Example Take

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & 3 \end{pmatrix}$$

Note that the fundamental object is the *linear map*, which upon choosing a basis happens to have a representation in terms of a matrix.

This is just a fancy way of saying composition of linear maps is represented by matrix multiplication. Note the order of operation matters.

to represent $\theta : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ with respect to the standard basis. Find A' that represents θ with respect to

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \right\} \quad \& \quad \left\{ \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}.$$

Since we are just transforming from the standard basis, we have

$$Y = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ -1 & 1 & 0 \end{pmatrix} \quad \& \quad X = \begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix},$$

and so

$$A' = XAY = \begin{pmatrix} -5 & 3 & -3 \\ 2 & 1 & 4 \end{pmatrix}.$$

The following are **elementary matrices** (those starting from the identity matrix essentially)

- multiply row by a constant c , e.g.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & c & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

- interchange a row, e.g.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

- linear addition of one row to another, e.g.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix}$$

For a matrix $A = (a_{ij})$, its **transpose** is $A^T = (a_{ji})$.

A matrix A is said to be **invertible** with **inverse** A^{-1} if

$$A^{-1}A = AA^{-1} = I. \quad (2.2)$$

Theorem 2.0.3 *If a matrix A is invertible then there is a product of elementary matrices E_i where $E_1 \cdots E_k A = I$.*

So then $A^{-1} = E_1 \cdots E_k$.

Proof Since A is invertible A does not contain a row of zeroes. By arranging rows accordingly and that there are no rows that are linear combinations of another exactly (otherwise there is a row of zeroes), we can do the elementary operations to get to I , and since each elementary operation corresponds to an elementary matrix multiplication, we have the above result. ■

Example Starting from

$$A = \begin{pmatrix} 3 & -3 & 1 \\ -4 & 5 & 2 \\ 2 & -2 & 1 \end{pmatrix},$$

we have

$$\begin{aligned} & \left(\begin{array}{ccc|ccc} 3 & -3 & 1 & 1 & 0 & 0 \\ -4 & 5 & -2 & 0 & 1 & 0 \\ 2 & -2 & 1 & 0 & 0 & 1 \end{array} \right) \\ & \rightarrow \left(\begin{array}{ccc|ccc} 1 & -1 & 0 & 1 & 0 & -1 \\ 0 & 1 & 0 & 0 & 1 & 2 \\ 2 & -2 & 1 & 0 & 0 & 1 \end{array} \right) \\ & \rightarrow \left(\begin{array}{ccc|ccc} 1 & -1 & 0 & 1 & 0 & -1 \\ 0 & 1 & 0 & 0 & 1 & 2 \\ 0 & 0 & 1 & -2 & 0 & 3 \end{array} \right) \\ & \rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 2 \\ 0 & 0 & 1 & -2 & 0 & 3 \end{array} \right), \end{aligned}$$

Trying to make as many 1s as possible at the desired spots, which then makes killing the entries above and below easier.

where we can read off the inverse A^{-1} .

2.1 Rank

Let A be a $m \times n$ matrix. Then the **row rank** of A is the dimension of the subspaces spanned by the row vectors of A . The **column rank** is defined similarly.

Theorem 2.1.1 *For every matrix, the row rank is equal to the column rank (so we only need to talk about rank).*

Proof Think of A as a linear map $\theta : \mathbb{R}^n \rightarrow \mathbb{R}^m$, so there exists non-singular X and Y where

$$B = XAY = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix},$$

where r is the row rank of θ . The transpose of B has the same rank, although that is now the column rank of B^T . ■

We define the **nullity** of A to be the dimension of the kernel, i.e., the dimension of the solution set $Ax = 0$.

Lemma 2.1.2 *The following are equivalent if A is a square $n \times n$ matrix:*

1. A is invertible,

2. $\text{rank}(A) = n$,
3. $\text{null}(A) = 0$.

Proof If A is invertible then the corresponding linear map θ is an isomorphism, and we have the result by Thm. 1.3.5. ■

Example Note that by elementary operations we have

$$A = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 2 & 2 & 3 & 2 \\ 3 & 3 & 5 & 4 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 2 & 4 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

so $r(A) = 2$.

2.2 Trace and determinant

Let $A = (a_{ij})$ be a $n \times n$ matrix. The **determinant** of A is defined as

$$|A| = \sum_{\sigma \in S_n} \text{sgn}(\sigma) a_{1\sigma(1)} a_{2\sigma(2)} \cdots a_{n\sigma(n)}, \quad (2.3)$$

where S_n is the permutation group of size n , and sgn is the **signature** of the permutation of σ where $\text{sgn}(\sigma) = \pm 1$ if σ is even or odd.

Lemma 2.2.1

$$|A| = |A^T|. \quad (2.4)$$

Proof We see that since $(a_{ij})^T = (a_{ji})$, for any $\sigma \in S_n$ and that $\text{sgn}(\sigma) = \text{sgn}(\sigma^{-1})$ and is just a number,

$$\begin{aligned} |A| &= \sum_{\sigma \in S_n} a_{\sigma^{-1}(1)1} \cdots a_{\sigma^{-1}(n)n} \text{sgn}(\sigma) \\ &= \sum_{\sigma \in S_n} a_{\sigma^{-1}(1)1} \cdots a_{\sigma^{-1}(n)n} \text{sgn}(\sigma^{-1}) \\ &= \sum_{\sigma \in S_n} \text{sgn}(\sigma^{-1}) a_{1\sigma^{-1}(1)}^T \cdots a_{n\sigma^{-1}(n)}^T, \end{aligned}$$

then we can just call $\tau = \sigma^{-1}$, and then we have the definition of $|A^T|$. ■

Lemma 2.2.2 • Multiplying a single row/column A by c multiplies $|A|$ by c .

- Permuting a single row/column of A with another multiplies $|A|$ by -1 .
- If two columns/rows are the same then $|A| = 0$.
- Adding multiples of a row/column to another does not change $|A|$.

• We have

$$\begin{vmatrix} b_{11} + c_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} + c_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} = \begin{vmatrix} b_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} + \begin{vmatrix} c_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix}$$

and similarly for rows. $\square$

Corollary 2.2.3 *If the rows of A are linearly dependent on row vectors then $|A| = 0$, and similarly for columns.*

Theorem 2.2.4

$$|AB| = |A||B|. \quad (2.5)$$

Proof If A is singular then $r(A) < n$, so that $r(AB) < n$ and so AB is also not invertible, so we have $0 = 0$, and analogously for the case where B is singular.

Otherwise A, B and AB are invertible, and they are combinations of elementary matrices, and by the above corollary we have what we need given $|EA| = |E||A|$. $\blacksquare$

Let A be a matrix, and let M_{ij} be the determinant obtained by deleting the i^{th} row and j^{th} column of A. M_{ij} are called the **minors** of A, and $A_{ij} = (-1)^{i+j}M_{ij}$ are the **cofactors** of A.

Lemma 2.2.5 *For fixed i or fixed j, we have*

$$|A| = \sum_{j=1}^n a_{ij}A_{ij} = \sum_{i=1}^n a_{ij}A_{ij},$$

i.e. the determinant of A can be expanded using a convenient row or column from the matrix of cofactors.

Example For

$$A = \begin{pmatrix} 1 & 0 & 2 \\ 3 & 4 & 5 \\ 0 & 1 & 0 \end{pmatrix},$$

expanding on the third row gives (neglecting the two terms multiplied by zero)

$$|A| = -1 \begin{vmatrix} 1 & 2 \\ 3 & 5 \end{vmatrix} = -(5 - 6) = 1,$$

while expanding on the second column gives (neglecting the first term multiplied by zero)

$$|A| = +4 \begin{vmatrix} 1 & 2 \\ 0 & 0 \end{vmatrix} - 1 \begin{vmatrix} 1 & 2 \\ 3 & 5 \end{vmatrix} = 0 - (5 - 6) = 1.$$

The **adjugate matrix** of A or the **adjoint** of A is the transpose of the matrix of cofactors of A , i.e.

$$(\text{adj}A)_{ij} = A_{ji}. \quad (2.6)$$

The adjoint is *almost* an inverse (see below), although an adjoint can be defined even when the inverse isn't.

Lemma 2.2.6 Let A be a $n \times n$ matrix, then $A(\text{adj}A) = |A|I = (\text{adj}A)A$.

Proof For fixed i , we have

$$|A| = \sum_{j=1}^n a_{ij}A_{ij} = \sum_{j=1}^n a_{ij}\text{adj}A_{ji} = (A\text{adj}A)_{ii},$$

where there is no sum implied on the last entry. For $i \neq k$, we have

$$(A\text{adj}A)_{ik} = \sum_{j=1}^n a_{ij}\text{adj}A_{jk} = \sum_{j=1}^n a_{ij}A_{kj}.$$

Let B be obtained by replacing the k^{th} row of A by the i^{th} row of A , so clearly we have

$$0 = |B| = \sum_{j=1}^n b_{kj}B_{kj} = \sum_{j=1}^n a_{ij}A_{kj},$$

which implies that

$$A\text{adj}A = |A|I$$

if $|A| \neq 0$. A similar manipulation for A^T leads to

$$|A^T|I = A^T\text{adj}A^T = A^T(\text{adj}A)^T = [(\text{adj}A)A]^T,$$

which gives our desired result. ■

Then clearly we have

$$A^{-1} = \frac{1}{|A|}\text{adj}A \quad (2.7)$$

if $|A| \neq 0$.

Theorem 2.2.7 A matrix A has an inverse iff $|A| \neq 0$.

Proof By previously lemma we have $|A| \neq 0$ implying existence of inverse. If there exists an inverse, then there is a sequence of elementary operations acting on A that results in I . By Thm. 2.2.4,

$$1 = |I| = |E_1 \cdots E_k A| = |E_1| \cdots |E_k| \cdot |A|,$$

and since $|E_i| \neq 0$, then $|A| \neq 0$. ■

Corollary 2.2.8 Let A be a square matrix. $|A| \neq 0$ iff row/column vectors form a basis of $\mathbb{R}^n$.

Proof $|A| \neq 0$ iff A invertible iff $r(A) = n$ iff row vectors form a basis. Since column and row rank are equal, the same is true for columns. ■

2.3 Eigenvalues and eigenvectors

Let V be a vector space, and $\theta : V \rightarrow V$ be a linear map. Then $v \in V - \{0\}$ is an **eigenvector** of θ if

$$\theta(v) = \lambda v, \quad (2.8)$$

where λ is the corresponding **eigenvalue**. The subset

$$V_\lambda = \{v \in V \mid \theta(v) = \lambda v\} \quad (2.9)$$

is a vector space called the **eigenspace** corresponding to the eigenvalue λ .

Clearly if A is the corresponding matrix to θ then the eigenvalues and eigenvectors would correspond accordingly.

Two $n \times n$ matrices A and B are called **similar** if there exists some non-singular Y where $B = Y^{-1}AY$.

Theorem 2.3.1 *If A corresponds to the linear map $\theta : V \rightarrow V$, then the roots of the **characteristic polynomial** $|\lambda I - A| = 0$ are the eigenvalues of θ . This is independent of the basis of V .*

Proof Let v be an eigenvector, so that $v = \sum_i v_i e_i$. If λ is the corresponding eigenvalue, and A is the representation of θ in the basis $\{e_i\}$, then

$$Av = \lambda v \Leftrightarrow (\lambda I - A)v = 0.$$

Since v is non-trivial by definition of it being an eigenvector, then $\lambda I - A$ is singular, so $|\lambda I - A| = 0$.

For basis independence, we know that

$$\begin{aligned} |\lambda I - Y^{-1}AY| &= |\lambda Y^{-1}IY - Y^{-1}AY| \\ &= |Y^{-1}(\lambda I - A)Y| \\ &= |Y^{-1}| \cdot |\lambda I - A| \cdot |Y|, \end{aligned}$$

but by the property that $|Y^{-1}| \cdot |Y| = 1$, we have $|\lambda I - Y^{-1}AY| = |\lambda I - A| = 0$. ■

Example For

$$A = \begin{pmatrix} 2 & 2 \\ 0 & 4 \end{pmatrix}, \quad \lambda I - A = \begin{pmatrix} \lambda - 2 & -2 \\ 0 & \lambda - 4 \end{pmatrix}$$

and the eigenvalues satisfy the quadratic $(\lambda - 2)(\lambda - 4) = 0$, so $\lambda = 2, 4$. For $\lambda = 2$, the eigenvector satisfies

$$\begin{pmatrix} 0 & -2 \\ 0 & -2 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = 0,$$

Could have done this with the trace and determinant also for a 2×2 matrix.

which implies $v_2 = 0$ but v_1 is undetermined; we could choose for simplicity for the eigenvector to be

$$\mathbf{v}_{\lambda=2} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

For $\lambda = 4$, we result in $v_1 = v_2$, so we could choose

$$\mathbf{v}_{\lambda=4} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Notice for the above that if we take $\mathbf{Y}$ to be the matrix whose column vectors are the eigenvectors, i.e.

$$\mathbf{Y} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \Rightarrow \mathbf{Y}^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix},$$

then notice that

$$\mathbf{Y}^{-1}\mathbf{A}\mathbf{Y} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 2 \\ 0 & 4 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix},$$

which is just the **diagonal** matrix with the eigenvalues on the main diagonal. This is called **diagonalisation**.

Theorem 2.3.2 Let $\mathbf{A}$ be a $n \times n$ matrix. The matrix $\mathbf{A}$ is **diagonalisable** if there exists a non-singular $\mathbf{Y}$ such that $\mathbf{A}$ is similar to a diagonal matrix $\mathbf{D}$, i.e. $\mathbf{D} = \mathbf{Y}^{-1}\mathbf{A}\mathbf{Y}$ or $\mathbf{A} = \mathbf{Y}\mathbf{D}\mathbf{Y}^{-1}$. This is equivalent to the condition that the eigenvectors of $\mathbf{A}$ has full span.

Notice we do not require $\mathbf{A}$ to be non-singular, nor that the eigenvalues have to be distinct. For example,

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

are respectively singular and have repeated eigenvalues, but are both trivially diagonal (so diagonalisable by $\mathbf{I}$). On the other hand,

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

has eigenvalues $\lambda = \pm i$, so it is definitely not diagonalisable over the reals.

Example For

$$\mathbf{A} = \begin{pmatrix} 3 & 0 \\ 1 & 3 \end{pmatrix},$$

we have $\lambda = 3$ of multiplicity 2, but additionally only one eigenvector $(1, 0)^T$, so the span of the eigenspace is only dimension 1, and matrix is not diagonalisable.

Example For

$$A = \begin{pmatrix} 0 & -2 & 2 \\ 0 & 2 & 0 \\ -1 & -1 & 3 \end{pmatrix},$$

can show that the eigenvalues are $\lambda = 1$ and $\lambda = 2$ twice. For eigenvector corresponding to $\lambda = 1$ can be chosen to be

$$v_1 = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}.$$

For $\lambda = 2$, we reduce down to $v_1 + v_2 - v_3 = 0$, so we could take

$$v_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad v_3 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix},$$

which are linearly independent of each other, so the eigenspace has full span, and the transformation matrix could be

$$Y = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}.$$

Notice we could have made a different choice for $v_{2,3}$, and that

$$Y = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 0 & 1 \\ 2 & -1 & 1 \end{pmatrix}$$

would have also worked.

There are at least two notable uses for this:

Example 1. If we have the system of ODEs

$$\frac{d}{dt}x = Ax$$

where A is constant and diagonalisable with $A = YDY^{-1}$, then the solutions are

$$x = Y \begin{pmatrix} e^{\lambda_1 t} \\ \vdots \\ e^{\lambda_n t} \end{pmatrix}$$

up to multiplicative constants (which are specified by the initial conditions). This is because

$$\frac{d}{dt}x = YDY^{-1}x \Rightarrow \frac{d}{dt}Y^{-1}x = DY^{-1}x,$$

because Y is constant in time so commutes with the derivative. Then calling $y = Y^{-1}x$, we just have a set of trivial constant ODEs to solve, then we invert to get $x = Yy$ which is known.

In other words, we choose a convenient basis where our problem is diagonal, solve the problem there (which is easy), and then transform it back into the original basis. This kind of problem occurs occasionally in quantum mechanics, where you might want to solve the **time-independent Schrödinger equation**

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H |\psi(t)\rangle$$

where H would be matrix representation of the relevant Hamiltonian of the system, and $|\psi(t)\rangle$ would be the vector representation of the **state**. The eigenvalues correspond usually to an energy, and the eigenvectors the energy states.

The Hamiltonian normally has additional properties that make the eigenvalues and eigenvectors have certain properties; more later when we talk about *Hermitian/self-adjoint* matrices.

2. Suppose we have some system that is governed by

$$x_{n+1} = Ax_n,$$

such as a **Markov chain** with discrete states, and so A is a matrix detailing the transition probabilities between states (with all positive entries that are strictly less than 1, since these are supposed to describe probabilities). If A is diagonalisable, then the numerics becomes 'easy', in that to find x_{n+1} from initial data x_0 , we do

$$\begin{aligned} x_n &= A^n x_0 \\ &= (YDY^{-1})^n x_0 \\ &= (YDY^{-1})(YDY^{-1}) \cdots (YDY^{-1}) x_0 \\ &= YD(Y^{-1}Y)D \cdots (Y^{-1}Y)DY^{-1} x_0 \\ &= YD^n Y^{-1} x_0, \end{aligned}$$

by associativity, and noting that D^n is essentially trivial to compute.

2.4 Cayley–Hamilton theorem

In the above we very loosely and briefly touched on the **matrix exponential**, which is

$$e^{At} = I + \frac{1}{1!}A + \frac{1}{2!}A^2 + \cdots = \sum_{k=0}^{\infty} \frac{1}{k!}A^k. \quad (2.10)$$

These crop up quite often with differential equations, and there is a link between the higher and lower powers of the matrix.

In the context of a Markov chain, the transition matrix have eigenvalues with magnitude less than or equal to 1 by the *Perron–Frobenius theorem*, so we have no blow up. This is not necessarily true for population models such as the *Leslie model*, since the entries are positive but do not have to be strictly less than 1.

Which relates also to the **exponential map** between a Lie group and its Lie algebra.

Theorem 2.4.1 Every $n \times n$ matrix A satisfies its own characteristic polynomial, i.e.

$$0 = A^n + a_{n-1}A^{n-1} + \dots a_1A + a_0I. \quad (2.11)$$

Proof Let $|tI - A| = t^n + a_{n-1}t^{n-1} + \dots + a_0$. Let $B(t) = \text{adj}(tI - A)$, whose entries are by definition the cofactors of $tI - A$, so

$$B(t) = B_{n-1}t^{n-1} + \dots B_1t + B_0.$$

By definition we have

$$I = (tI - A) \left[\frac{1}{|tI - A|} \text{adj}(tI - A) \right] \Rightarrow I|tI - A| = (tI - A)B,$$

we have

$$(tI - A)(B_{n-1}t^{n-1} + \dots B_1t + B_0) = (t^n + a_{n-1}t^{n-1} + \dots + a_0)I.$$

Comparing coefficients we have $B_{i-1} - AB_i = a_{i-1}I$ accounting for edge cases appropriately. Multiplying both sides by A^i we have $A^iB_{i-1} - A^{i+1}B_i = a_{i-1}A^i$, and summing both sides accounting for edge cases appropriately gives

$$A^nB_{n-1} + \left(\sum_{i=1}^{n-1} A^iB_{i-1} - A^{i+1}B_i \right) - AB_0 = A^n + \sum_{i=0}^{n-1} a_iA^i.$$

The left hand side cancels out exactly, and we have the result. ■

So solving the eigenvalue problem associated with A gives the a_k s, and then you can get the higher powers from lower powers recursively with matrix addition rather than matrix multiplication, where the latter has higher computational complexity.

2.5 Orthogonal matrices

Recall that two vectors are orthogonal to each other if $u \cdot v = u^T v = 0$, the latter interpreted in terms of a matrix multiplication of a row vector with a column vector.

Analogously, a matrix is called **orthogonal** if the columns are orthogonal to each other. However, a more useful form of orthogonality might be the following:

Theorem 2.5.1 A matrix A is orthogonal if and only if $A^{-1} = A^T$.

Proof Let e_i be the i^{th} column of P , then $AA^T = (e_i^T e_j) = (\delta_{ij}) = I$, as required. ■

So orthogonal matrices are great, because a transpose is much easier to compute than the inverse.

2.6 Symmetric matrices

A matrix A is **symmetric** if $A = A^T$.

Theorem 2.6.1 Let A be a real symmetric matrix. Then the eigenvalues are all real, and the eigenvectors are mutual orthogonal to each other. $\square$

Theorem 2.6.2 Let A be a real symmetric matrix. If it is diagonalisable, then the similarity matrix is orthogonal, i.e. $A = PDP^{-1} = PDP^T$. $\square$

The theorems are special cases of those given in §2.8, so we defer the proof for later.

Example Diagonalise

$$A = \begin{pmatrix} -1 & 1 & -1 \\ 1 & -1 & -1 \\ -1 & -1 & -1 \end{pmatrix}.$$

Can show the eigenvalues are 1, -2 and -2 , and that

$$P = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & -2 \\ -1 & 1 & 1 \end{pmatrix}$$

is one possible choice, leading to $D = \text{diag}(1, -2, -2)$.

2.7 Rotations

Denote the set of $n \times n$ orthogonal matrices over $\mathbb{R}$ by $O(n)$, and $SO(n)$ the subset of $O(n)$ where the corresponding matrices have determinant $+1$. These turn out to form a group under matrix multiplication, and are known as the **orthogonal** and **special orthogonal** groups of dimension n respectively.

Theorem 2.7.1 Every element of $SO(3)$ represent a rotation about an arbitrary axis in $\mathbb{R}^3$. $\square$

Corollary 2.7.2 The characteristic polynomial

$$\Delta(\lambda) = \lambda^3 - (1 + 2 \cos \theta)\lambda^2 + (1 + 2 \cos \theta)\lambda - 1$$

tells us the angle of rotation θ (via $\cos \theta$, up to a sign) of a rotation about any axis.

Example Find the matrix which represents a rotation in $\mathbb{R}^3$ about $(1, 1, 0)^T$ through an angle of $\pi/2$.

Note that we have

$$R_x(\theta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{pmatrix}, \quad R_y(\theta) = \begin{pmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{pmatrix},$$

$$R_z(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

These are actually **Lie groups**, which play an important role in mathematical physics.

are the rotation matrices about the x , y and z axis respectively by an angle θ . Then idea is that the axis is represented by a vector that we can normalise, and we rotate that onto a standard axis (essentially a co-ordinate transformation), do the rotation by the desired angle (which we know the matrix of), and then undo rotation (or the co-ordinate transformation).

We can take the z axis for definiteness, and we want $P : (1, 1, 0)^T / \sqrt{2} \mapsto (0, 0, 1)^T$. This is supposed to be a rotation, so P^{-1} exists and $P^{-1} : (0, 0, 1)^T \mapsto (1, 1, 0)^T / \sqrt{2}$, or

$$P^{-1} = \begin{pmatrix} \cdot & \cdot & 1/\sqrt{2} \\ \cdot & \cdot & 1/\sqrt{2} \\ \cdot & \cdot & 0 \end{pmatrix}.$$

The column vectors should be mutually orthogonal to each other, and we can choose them to be orthonormal (orthogonal vectors that are normalised to unit length). We could consider the simple choice

$$P^{-1} \stackrel{!}{=} \begin{pmatrix} 0 & -1/\sqrt{2} & 1/\sqrt{2} \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \\ 1 & 0 & 0 \end{pmatrix}.$$

Notice however that $|P^{-1}| = -1$, so to make sure $P \in \text{SO}(3)$, we can simply multiply the bottom row by -1 . The resulting P is then

$$P = \begin{pmatrix} 0 & 1/\sqrt{2} & 1/\sqrt{2} \\ 0 & -1/\sqrt{2} & 1/\sqrt{2} \\ 1 & 0 & 0 \end{pmatrix},$$

and so

$$A = P^{-1}R_z(\pi/2)P = \begin{pmatrix} 1/2 & 1/2 & 1/\sqrt{2} \\ 1/2 & 1/2 & -1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} & 0 \end{pmatrix}$$

will do the job. Can brute force check that that $|A| = 1$, $A^T = A$, and the only solution to $Ax = x$ is $x = a(1, 1, 0)^T$.

The rotation axis is kept fixed by the rotation about that axis.

2.8 Hermitian matrices

A matrix over $\mathbb{C}$ is **Hermitian** if $A = A^\dagger = \overline{A}^T$. Note that if A is actually real, then this reduces to the definition of a symmetric matrix.

A matrix over $\mathbb{C}$ is **unitary** if $A^\dagger = A^{-1}$. Again, if A is actually real, then this reduces to the definition of an orthogonal matrix.

Unitary matrices are used a lot in quantum mechanics.

Theorem 2.8.1 *Let A be a Hermitian matrix. Then the eigenvalues are all real, and the eigenvectors are mutual orthogonal to each other.*

Compare this with Theorem 2.6.1. In quantum mechanics the Hamiltonians are usually (but not always) chosen to be Hermitian, so the eigenvalues are all real and observable.

Proof Let λ be an eigenvalue of A , so $Av = \lambda v$ with $v \neq \mathbf{0}$. Now,

$$\lambda \|v\|^2 = \lambda v^\dagger v = v^\dagger (\lambda v) = v^\dagger Av = v^\dagger A^\dagger v = (Av)^\dagger v = \bar{\lambda} v^\dagger v = \bar{\lambda} \|v\|^2,$$

so λ is real for non-zero v .

Let λ and μ be distinct eigenvalues of A , so there exists associated eigenvectors v and w respectively. Then doing similar manipulations to above,

$$\lambda w^\dagger v = w^\dagger Av = (Aw)^\dagger v = \mu w^\dagger v.$$

Since $\lambda \neq \mu$, then $w^\dagger v = \langle v, w \rangle = 0$, and so the eigenvectors are orthogonal. ■

Quadratic forms show up areas like number theory, physics and differential geometry (the Riemannian metric, the second fundamental form, Killing form) and statistics.

A **quadratic form** Q on a real inner product space V is a function $Q : V \rightarrow \mathbb{R}$ of the form $Q(v) = \langle Av, v \rangle$, where A is some symmetric matrix. Note that this is a polynomial of degree two. If $\{e_1, \dots, e_n\}$ is a basis of B with $v = x_i e_i$, then

$$Q(x) = x^T A x = \sum_{i=1}^n a_{ii} x_i^2 + 2 \sum_{1 \leq i < j \leq n} a_{ij} x_i x_j.$$

A **symmetric bilinear form** B on a vector space V is a function $B : V \times V \rightarrow \mathbb{R}$ such that $B(v, w) = B(w, v)$, and B is linear in the first factor.

A **Hermitian form** H on a complex inner product space V is a function $H : V \rightarrow \mathbb{R}$ of the form $H(v) = \langle Av, v \rangle$ where A is Hermitian ($A = A^\dagger$).

If $\{e_1, \dots, e_n\}$ is a basis of V and $v = z_i e_i$, then

$$H(v) = v^\dagger A v = \sum_{i=1}^n a_{ii} |v_i|^2 + \sum_{i < j} (a_{ij} \bar{z}_i z_j + a_{ji} \bar{z}_j z_i).$$

A **symmetric sesquilinear form** S on a complex vector space V is $S : V \times V \rightarrow \mathbb{C}$ where $S(w, v) = \overline{S(v, w)}$, and S is linear in the first factor.

A symmetric bilinear form defines a quadratic form via $Q(v) = B(v, v)$. Conversely, Q determines B via $B(v, w) = (1/2)(Q(v + w) - Q(v) - Q(w))$, and that

$$B(x, y) = y^T A x \quad \Leftrightarrow \quad Q(x) = x^T A x.$$

A bilinear form B is called **positive definite** if $B(v, v) = Q(v) > 0$ and $(v, v) = H(v) > 0$ for all $v \neq 0$.

Lemma 3.0.1 *A symmetric bilinear/sesquilinear form is an inner product iff it is positive.* □

A symmetric or Hermitian matrix is **positive definite** iff the bilinear form $B(x, y) = y^T A x$ is positive definite.

Theorem 3.0.2 *A symmetric/Hermitian matrix A is positive definite iff all eigenvalue are positive.*

Proof Let $H(z) = z^T A z$. There is an orthonormal basis $\{u_1, \dots, u_n\}$ so that $H(z_i u_i) = \lambda_i |z_i|^2$ where λ_k are the eigenvalues.

If A is positive definite then acting on the orthonormal basis $H(u_i) = \lambda_i > 0$ for all i , so all eigenvalues are positive.

If all eigenvalues are positive, then $\lambda_i |z_i|^2 > 0$ for all i and so $H(z) > 0$, so it is positive definite. ■

Theorem 3.0.3 *An square symmetric matrix A is positive definite iff its principal minors all have positive determinant.* □

Example For

$$A = \begin{pmatrix} 2 & 2 & -2 \\ 2 & 5 & -4 \\ -2 & -4 & 5 \end{pmatrix},$$

it can be shown that A is positive definite. The principal minors are

$$|2| = 2, \quad \begin{vmatrix} 2 & 2 \\ 2 & 5 \end{vmatrix} = 6, \quad \begin{vmatrix} 2 & 2 & -2 \\ 2 & 5 & -4 \\ -2 & -4 & 5 \end{vmatrix} = 10.$$

Lemma 3.0.4 *Let A be a complex Hermitian matrix, then all eigenvalues of A are real, and all eigenvectors of A are mutually orthogonal.*

This is partly why in quantum mechanics normally deals with Hermitian matrices.

Proof Let λ be an eigenvalue of A, so $Ax = \lambda x$ for $x \neq 0$. Then

$$\lambda \|x\|^2 = x^\dagger (\lambda x) = x^\dagger A x = x^\dagger A^\dagger x = (Ax)^\dagger x = \bar{\lambda} x^\dagger x = \bar{\lambda} \|x\|^2,$$

and since $x \neq 0$, $\lambda \in \mathbb{R}$.

Let λ and μ be distinct eigenvalues of A, so there exists non-zero x and y that $Ax = \lambda x$ and $Ay = \mu y$. Then

$$\lambda y^\dagger x = y^\dagger A x = (Ay)^\dagger x = \mu y^\dagger x,$$

but $y^\dagger x = \langle y, x \rangle$, and since $\lambda \neq \mu$, we have $\langle y, x \rangle = 0$, and we have orthogonal eigenvectors. ■

Theorem 3.0.5 *Let A be a Hermitian matrix. If it is diagonalisable, then the similarity matrix is unitary, i.e. $A = PDP^{-1} = PDP^\dagger$.*

Proof Let λ_1 be an eigenvalue of A with eigenvector v . Let $U = \text{span}\{v\}$. For $w \in U^\perp$, then

$$\langle v, Aw \rangle = \langle Av, w \rangle = \langle \lambda_1 v, w \rangle = \lambda_1 \langle v, w \rangle = 0,$$

hence $Aw \in U^\perp$. Supposing $\|v\| = 1$, we choose an orthonormal basis $u_2, \dots, u_n$ for $U^\perp$, which implies that $\{v, u_2, \dots, u_n\}$ is an orthonormal basis for $\mathbb{C}^n$.

Let P_1 be the matrix whose columns are those vectors, which by construction means $P_1^\dagger P_1 = I$ by orthogonality. Let $\{e_i\}$ be the standard basis of $\mathbb{C}^n$, then

$$P_1^\dagger A P_1 e_1 = P_1^\dagger A v = P_1^\dagger \lambda_1 v = \lambda_1 P_1^\dagger v = \lambda_1 e_1 \quad \Rightarrow \quad P_1^\dagger A P_1 = \begin{pmatrix} \lambda_1 & 0 \\ 0 & A_1 \end{pmatrix}.$$

The procedure can be continued for the next one, so that

$$P_2^\dagger P_1^\dagger A P_1 P_2 = \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & A_2 \end{pmatrix},$$

and so we can define $P = \prod_{i=1}^n P_i$ as the transformation matrix. Since each P_i is unitary, its product is unitary. ■

Example Diagonalise the real symmetric matrix (and therefore Hermitian)

$$A = \begin{pmatrix} 2 & -2 & 0 \\ -2 & 1 & -2 \\ 0 & -2 & 0 \end{pmatrix}.$$

The characteristic polynomial is $|\lambda I - A| = (\lambda - 1)(\lambda - 4)(\lambda + 2) = 0$ so the eigenvalues are $\lambda = 1, 4, -2$, with associated unit eigenvectors

$$\frac{1}{3} \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}, \quad \frac{1}{3} \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}, \quad \frac{1}{3} \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix},$$

so

$$P = \frac{1}{3} \begin{pmatrix} 2 & 2 & 1 \\ -2 & 1 & 2 \\ 1 & -2 & 2 \end{pmatrix}, \quad P^T A P = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}.$$

Example Diagonalise the Hermitian matrix

$$A = \begin{pmatrix} 4 & 1-i \\ 1+i & 5 \end{pmatrix}.$$

The characteristic polynomial is $|\lambda I - A| = (\lambda - 3)(\lambda - 6) = 0$ so the eigenvalues are $\lambda = 3, 6$ and real (as they should be). The associated unit eigenvectors are

$$\frac{1}{\sqrt{3}} \begin{pmatrix} -1+i \\ 1 \end{pmatrix}, \quad \frac{1}{\sqrt{6}} \begin{pmatrix} 1-i \\ 2 \end{pmatrix},$$

so

$$P = \begin{pmatrix} (-1+i)/\sqrt{3} & (1-i)/\sqrt{6} \\ 1/\sqrt{3} & 2/\sqrt{6} \end{pmatrix}, \quad P^T A P = \begin{pmatrix} 3 & 0 \\ 0 & 6 \end{pmatrix}.$$

Theorem 3.0.6 (Simultaneous diagonalisation) Let A and B be symmetric matrices and B be positive definite. There exists some $|P|$ where $|P| \neq 0$ and $P^T B P = I$ and $P^T A P = D$ where D is a real diagonal matrix. Moreover, entries of D are the roots of $|\lambda B - A| = 0$ and the columns of P form an orthonormal basis with respect to the inner product defined by B .

Proof As B is symmetric and positive definite (so Hermitian), there exists some orthogonal (a real unitary) matrix P_0 so that

$$P_0 B P_0 = D_0 = \text{diag}(\lambda_1, \dots, \lambda_n).$$

1. Let $D_1 = \text{diag}(1/\sqrt{\lambda_1}, \dots, 1/\sqrt{\lambda_n})$, so that $D_1^T P_0^T B P_0 D_1 = I$ by construction, so we can define $P_1 = P_0 D_1$ and we have $P_1^T B P_1 = I$. Note here that P_1 is not orthogonal, and this only makes sense if B is positive definite (otherwise $1/\sqrt{\lambda_k}$ doesn't make sense).

Let $C = P_1^T A P_1$, and note that C is symmetric, so there exists P_2 where $P_2^T C P_2 = D$ where P_2 is orthogonal and D is a real diagonal matrix. Then we let $P = P_1 P_2$ and we would have $P^T A P = P_2^T D P_2 = D$ since P_2 is orthogonal. We also have $P^T B P = P_2^T I P_2 = I$ for the same reason.

2. The columns of P_2 are unit eigenvectors of C . Let v_i be one of these columns, and λ_i the corresponding eigenvalue that is also an entry of D . Then $P e_i = P_1 P_2 e_i = P_1 v_i$ since multiplying by the standard basis just picks out the relevant column.

Now, $P_1^T A P_1 v_i = C v_i = \lambda_i v_i$ by definition of C and that v_i is an associated eigenvector. Since $P_1^T B P_1 = I$, we have

$$P_1^T A P_1 v_i = \lambda_i (P_1^T B P_1) v_i,$$

but then this factorises to

$$P_1^T (\lambda_i B - A) P_1 v_i = 0,$$

and since $v_i \neq 0$ and $|P_1| \neq 0$, we have $|\lambda_i B - A| = 0$ as stated.

3. If y_i and y_j are columns of P then $y_j^T B y_i = \delta_{ij}$ since this is the $(ij)^{\text{th}}$ of $P^T B P = I$. So columns form an orthonormal basis as they are also unit vectors. ■

Example Find a non-singular P that simultaneously diagonalises

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix}.$$

Note that $|A| = -3$ but $|B| = 1$, so B is a positive definite matrix. Then $|\lambda B - A| = (\lambda + 3)(\lambda - 1) = 0$ so $\lambda = -3, 1$. We can check that the eigenvectors are

$$v_{\lambda=1} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad v_{\lambda=-3} = \begin{pmatrix} 1 \\ -2 \end{pmatrix},$$

and that $\|v_{\lambda=1}\|_B^2 = v_{\lambda=1}^T B v_{\lambda=1} = 1$ and similarly for $v_{\lambda=-3}$, so they are unit vectors with respect to the inner product defined by B . Then

$$P = \begin{pmatrix} 0 & 1 \\ 1 & -2 \end{pmatrix}$$

and that

$$P^T A P = \begin{pmatrix} 1 & 0 \\ 0 & -3 \end{pmatrix}, \quad P^T B P = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Theorem 3.0.7 Any quadratic form on $\mathbb{R}^n$ may be written as

$$Q(x) = x_1^2 + \dots + x_p^2 - x_{p+1}^2 - \dots - x_r^2$$

for $0 \leq o \leq r \leq n$ after a suitable linear change.

Proof If with respect to the standard co-ordinates we have $Q(x) = x^T A x$ with $A = A^T$ (since this is a quadratic form so A is symmetric), then there exists some orthogonal P that diagonalises A . We then define $y = P^T x$ so that

$$Q(x) = x^T A x = (P y)^T A (P y) = y^T D y = \lambda_k y_k^2$$

where λ_k are the eigenvalues of A . Suppose we rearrange accordingly so that $\lambda_i > 0$ for $1 \leq i \leq p$, $\lambda_i < 0$ for $p+1 \leq i \leq r$ and $\lambda_i = 0$ for $r+1 \leq i \leq n$. Then we can set $z_i = \sqrt{\lambda_i} y_i$, $z_i = \sqrt{-\lambda_i} y_i$ and $z_i = y_i$ for the respective cases so obtain $Q(z)$ in the desired form. ■

In the above r and $s = p - (r - p) = 2p - r$ is called the **rank** and **signature** of Q . Note that the parity of the rank and signature would be the same by construction. The rank is clearly related to the matrix representing the quadratic form, while the signature relates to the trace of the matrix after it has been transformed.

Example For $Q(x) = 3x_1^2 - x_2^2 + 4x_3^2 - 6x_1x_2 + 16x_2x_3$, note that

$$\begin{aligned} Q(x) &= 3(x_1 - x_2)^2 - 4x_2^2 + 4x_3^2 + 16x_2x_3 \\ &= 3(x_1 - x_2)^2 - 4(x_2 - 2x_3)^2 + 20x_3^2, \end{aligned}$$

then we can read off $tr = 3$ and $s = 2 - (3 - 2) = 1$.

Or there are two pluses and one minus, so signature is $1 + 1 - 1 = 2$. Compare this with for example when we say the Minkowski metric in special relativity having a signature of $(-, +, +, +)$ or $+2$ (or having the signs flipped depending on convention).

Let Q be a quadratic form on $\mathbb{R}^n$. A choice of co-ordinates $y_1, \dots, y_n$ on $\mathbb{R}^n$ is said to **diagonalise** Q if, with respect to these co-ordinates,

$$Q(\mathbf{y}) = d_k y_k^2 = \mathbf{y}^T \text{diag}(d_1, \dots, d_n) \mathbf{y}.$$

A pair of quadratic forms Q_1 and Q_2 are **simultaneously diagonalisable** if there is a choice of co-ordinates $y_1, \dots, y_n$ on $\mathbb{R}^n$ where $Q_1 = d_k y_k^2$ and $Q_2 = d'_k y_k^2$. Equivalently, A and B representing Q_1 and Q_2 with respect to the standard basis are simultaneously diagonalisable. Note that this is not always possible.

Theorem 3.0.8 (Sylvester's principle of inertia) *The rank and signature of a quadratic form is independent of how it is diagonalised provided it is done in the most efficient way.*

Proof The rank is clearly invariant by construction. Suppose we have two representations of Q

$$\begin{aligned} Q(\mathbf{x}) &= x_1^2 + \dots + x_p^2 - x_{p+1}^2 - \dots - x_r^2, \\ Q(\mathbf{y}) &= y_1^2 + \dots + y_q^2 - x_{q+1}^2 - \dots - y_r^2 \end{aligned}$$

where without loss of generality we assume $q < p$, and we aim to get a contradiction.

Let V_1 be the subspace of $\mathbb{R}^n$ spanned by $\{x_1, \dots, x_p\}$ (the positive parts), and it has dimensions $p + (n - r)$, with $Q(\mathbf{x}) \geq 0$ for $\mathbf{x} \in V_1$. However, let V_2 be the subspace spanned by $\{y_{q+1}, \dots, y_r\}$ (the negative parts), so that has dimensions $r - q$ with $Q(\mathbf{y}) \leq 0$ for $\mathbf{y} \in V_2$. Now, by construction, $\mathbf{x} \neq \mathbf{0}$ but also $\mathbf{x} \in V_1 \cap V_2$ because we assumed $q < r$, but then this implies $Q(\mathbf{x}) \geq 0$ and $Q(\mathbf{x}) \leq 0$, so $V_1 \cap V_2 = \{\mathbf{0}\}$ implying $\dim(V_1) + \dim(V_2) = n$. But above construction implies also $\dim(V_1) + \dim(V_2) = p + (n - r) + (r - q) = n + p - q > n$ since $p > q$, so we have a contradiction, implying $p = q$, so the signature should also be the same. ■

Theorem 3.0.9 *If Q_1 and Q_2 are quadratic forms represented by A_1 and A_2 , then there A_1 and A_2 are **congruent** (i.e. $A_2 = P^T A_1 P$ for some orthogonal P) iff Q_1 and Q_2 have the same rank and signature.*

Recall two matrices are *similar* if $A_2 = P^{-1} A_1 P$. Congruence is an equivalence relation.

Proof P above is just a co-ordinate transformation on $\mathbb{R}^n$ so Q_1 and Q_2 are the same quadratic forms in different co-ordinates, so by Sylvester's principle they have the same rank and signature.

For the converse, A_1 and A_2 are symmetric so there exists orthogonal P_1 and P_2 where

$$P_1^T A_1 P_1 = \text{diag}(1, 1, \dots, -1, -1, \dots, 0, \dots) = P_2^T A_2 P_2$$

since Q_1 and Q_2 have the same rank and signature. Then

$$A_2 = (P_2^T)^{-1} P_1^T A_1 P_1 P_2^{-1} = (P_1 P_2^{-1})^T A_1 (P_1 P_2^{-1}) = P^T A_1 P$$

as required (product of two orthogonal matrices is also orthogonal).

■

4 More on vector spaces and linear maps

Vector spaces and linear maps are however more general (e.g. maps on functions in function spaces), and here we look at other properties of vector spaces and linear maps in the more abstract setting.

4.1 Spaces of linear maps

The **module homomorphisms** $\text{Hom}(U, V)$ of two vector spaces U and V consists of all linear maps from U to V such that

$$(f + g)(u) = f(u) + g(u), \quad (\lambda f)(u) = \lambda f(u)$$

for all $u \in U, \lambda \in \mathbb{F}$.

Theorem 4.1.1 *$\text{Hom}(U, V)$ is a vector space over $\mathbb{F}$ and essentially follows from definition.* ■

Theorem 4.1.2 *Let $\dim(U) = m$ and $\dim(V) = n$, and the respective basis are $\{e_1, \dots, e_m\}$ and $\{f_1, \dots, f_n\}$. Let*

$$\theta_{ij}(e_k) = \delta_{ik} f_j$$

where δ_{ik} is the Kronecker delta. Then $\{\theta_{ij} \in \text{Hom}(U, V) \mid 1 \leq i \leq m, 1 \leq j \leq n\}$ is a basis of $\text{Hom}(U, V)$, and so $\dim(\text{Hom}(U, V)) = mn$.

Proof By Lemma 1.3.6 $\theta_{ij}(e_k) = \delta_{ik} f_j$ defines a unique map $\theta_{ij} : U \rightarrow V$. Suppose $\lambda_{ij} \theta_{ij} = 0$, then for all k where $1 \leq k \leq m$, $\lambda_{ij} \theta_{ij}(e_k) = \lambda_{ij} \delta_{ik} f_j = 0$. Since f_j is a basis of V , so for all j where $1 \leq j \leq n$ we have $\lambda_{ij} \delta_{ik} = 0$ implies that $\lambda_{kj} = 0$, so we have a set that is linearly independent.

Suppose $\phi \in \text{Hom}(U, V)$, then $\phi(e_k) = a_{kj} f_j$ for $a_{kj} \in \mathbb{F}$. But we also have $a_{ij} \theta_{ij}(e_k) = a_{ij} \delta_{ik} f_j = a_{kj} f_j$, so $\phi = a_{ij} \theta_{ij}$, and so θ_{ij} is a basis and $\text{Hom}(U, V)$ has dimension mn . ■

4.2 Dual space

Let V be a vector space over $\mathbb{F}$. The space $V^* = \text{Hom}(V, \mathbb{F})$ is called **dual** of V .

The elements of the dual here are maps from vectors to a field element (e.g. scalars).

Theorem 4.2.1 Let V be a vector space of dimension n over $\mathbb{F}$ and $\{e_1, \dots, e_n\}$ be a basis of V . Then there is a **dual basis** $\{e_1^*, \dots, e_n^*\}$ of V^* where

$$e_i^*(e_j) = \delta_{ij}. \quad (4.1)$$

This is just a special case of Thm. 4.1.2. ■

Given an inner product we denote $\langle f, v \rangle = f(v)$, where $f^* \in V^*$ and $v \in V$ and satisfies

$$\langle f + g, v \rangle = f(v) + g(v), \quad \langle \lambda f, v \rangle = \lambda f(v).$$

Theorem 4.2.2 There exists a canonical isomorphism between V and V^{**} if V is finite dimensional.

Proof Suppose $\theta : V \rightarrow V^{**}$ with $\langle \theta(v), f \rangle = \langle f, v \rangle$ for all $v \in V$, $f \in V^*$. Then

$$\begin{aligned} \langle \theta(u + v), f, f \rangle &= \langle f, u + v \rangle \\ &= \langle f, u \rangle + \langle f, v \rangle \\ &= \langle \theta(u), f \rangle + \langle \theta(v), f \rangle \\ &= \langle \theta(u) + \theta(v), f \rangle. \end{aligned}$$

Similarly,

$$\begin{aligned} \langle \theta(\lambda v), f, f \rangle &= \langle f, \lambda v \rangle \\ &= \lambda \langle f, v \rangle \\ &= \lambda \langle \theta(v), f \rangle \\ &= \langle \lambda \theta(v), f \rangle, \end{aligned}$$

so θ is a linear map.

If $v \in \text{Ker}(\theta)$ then $\langle f, v \rangle = \langle \theta(v), f \rangle = \langle \theta, f \rangle = 0$ for all $f \in V^*$, but this implies $v = 0$, so θ is injective. Since $\dim(V) = \dim(V^*) = \dim(V^{**})$ is finite and θ is injective, θ is bijective, and we thus have an isomorphism. ■

Theorem 4.2.3 Let $\theta : U \rightarrow V$ be a linear map of vector spaces. Then $\langle \theta^*(v^*), u \rangle = \langle v^*, \theta(u) \rangle$ defines a linear map $\theta^* : V^* \rightarrow U^*$.

Proof Note we have

$$\begin{aligned} \langle \theta^*(\lambda v^* + \mu w^*), u \rangle &= \langle \lambda v^* + \mu w^*, \theta(u) \rangle \\ &= \lambda \langle v^*, \theta(u) \rangle + \mu \langle w^*, \theta(u) \rangle \\ &= \lambda \langle \theta^*(v^*), u \rangle + \mu \langle \theta^*(w^*), u \rangle \\ &= \langle \lambda \theta^*(v^*) + \mu \theta^*(w^*), u \rangle, \end{aligned}$$

so θ^* is a linear map as required. ■

This might be a little confusing: f is a function that is also an element of a vector space and is not bolded.

A concrete example is that a gradient of a function ∇g is really a dual vector (a **1-form**), although we normally view it as a vector (which we can when we deal with flat Euclidean space). Contraction with a vector as $\langle \nabla g, v \rangle$ gives the *magnitude* (a scalar) of the derivative of g in the direction of v .

Essentially the dual map is defined by how the original moves from one side of the inner product to the other (cf. the adjoint).

Theorem 4.2.4 Let $\theta : U \rightarrow V$ be a linear map, and $\dim(U) = m$ and $\dim(V) = n$ are finite. If we have a matrix A that represents θ with respect to basis $\{e_1, \dots, e_m\}$ of U and $\{f_1, \dots, f_n\}$ of V , then A^T represents θ^* with respect to basis $\{e_1^*, \dots, e_m^*\}$ of U^* and $\{f_1^*, \dots, f_n^*\}$ of V^* .

Proof By convention, $\theta(e_j) = a_{ij}f_i$, and we have

$$\langle \theta^*(f_k^*), e_j \rangle = \langle f_k^*, \theta(e_j) \rangle = \langle f_k^*, a_{ij}f_i \rangle = a_{ij} \langle f_k^*, f_i \rangle = a_{ij} \delta_{ki}.$$

If θ^* is represented by B , then

$$\langle \theta^*(f_k^*), e_j \rangle = \langle b_{ik}e_i^*, \theta(e_j) \rangle = b_{ik} \langle e_i^*, e_j \rangle = b_{ik} \delta_{ij},$$

which implies $a_{kj} = b_{jk}$, so $A^T = B$. ■

Example Let $V = \mathbb{R}^3$ and $x = (x, y, z) \in V$. Show that all $f \in V^*$ has the form $f(x) = ax + by + cz$ for $a, b, c \in \mathbb{R}$.

If we let $a = f(e_1)$, $b = f(e_2)$ and $c = f(e_3)$ (i.e. basically the standard dual basis), then by linearity

$$f(x) = xf(e_1) + yf(e_2) + zf(e_3) = ax + by + cz.$$

If instead of the standard basis we take

$$f_1 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \quad f_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad f_3 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix},$$

then the dual basis are maps to $\mathbb{R}$. Let $f_1^*(x) = ax + by + cz$, then since $f_i^*(f_j) = \delta_{ij}$ by definition, we have the system

$$\begin{aligned} 1 &= f_1^*(f_1) = c \Rightarrow c = 1 \\ 0 &= f_1^*(f_2) = a + c \Rightarrow a = -1 \\ 0 &= f_1^*(f_3) = a - b \Rightarrow b = -1 \end{aligned}$$

so that $f_1^*(x) = -x + y + z$. Continuing on, it can be shown that $f_2^*(x) = x + y$ and $f_3^* = -y$, and these can be checked to be linearly independent and spanning $\mathbb{R}$, and we have the dual basis of V^* given the choice of a basis in V .

4.3 Symmetric and self-adjoint maps

Let V be an inner product space, and $T : V \rightarrow V$ a linear map. Then T is **symmetric** or **self-adjoint** if

$$\langle T(v), w \rangle = \langle v, T(w) \rangle.$$

The linear map T is an **isometry** if

$$\langle T(v), T(w) \rangle = \langle v, w \rangle,$$

Sometimes we use the dagger to denote the adjoint, so in the self-adjoint case $T^\dagger = T$. The fact that Hermitian transposes also use the dagger symbol is not a coincidence, as we shall see shortly.

i.e. inner products are invariant.

Note that if $\{e_1, \dots, e_n\}$ is a basis of V , then T can be represented by $A = T(e_1 | \dots | T(e_n))$. Further, if the basis is orthonormal, then $\langle v, w \rangle = w^\dagger v = w^\dagger A v$.

Lemma 4.3.1 *Symmetric/self-adjoint and isometric linear maps can be represented by symmetric/Hermitian and orthogonal/unitary matrices respectively.*

Proof Let A represent a self-adjoint T , then

$$w^\dagger (Av) = w^\dagger (Tv) = \langle T(v), w \rangle = \langle v, T(w) \rangle = (T(w))^\dagger v = (Aw)^\dagger v = w^\dagger A^\dagger v \quad (4.2)$$

so that $A = A^\dagger$, i.e. a Hermitian matrix. Similarly, let A represent an isometry T , then

$$w^T v = \langle v, w \rangle = \langle T(v), T(w) \rangle = \langle Av, Aw \rangle = (Aw)^T (Av) = w^T A^T A v, \quad (4.3)$$

which implies $A^T A = I$, so A is unitary. The real case follows from the complex case. ■

Self-adjoint or Hermitian operators play quite a prominent role in quantum mechanics, since the eigenvalues (related to observables) are guaranteed to be real.

4.4 Reflections

This part utilises orthogonal projections from earlier. Let V be an inner product space and U a subspace of V . Suppose $v = u + \tilde{u}$ where $v \in V$ and $\tilde{u} \in U^\perp$. Then a **reflection** of V in U is a linear map

$$R : V \rightarrow V^\perp, \quad R(v) = u - \tilde{u}. \quad (4.4)$$

In particular, if U or $U^\perp$ are finite dimensional, then we can write

$$R(v) = 2 \langle v, u_i \rangle u_i - v = v - 2 \langle v, n_j \rangle n_j,$$

where $\{u_1, \dots, u_k\}$ is a basis of U and $\{n_1, \dots, n_l\}$ is a basis of $U^\perp$.

Example Find the image of $(1, 2, 3)$ under the reflection in the plane $2x_1 + 2x_2 - x_3 = 0$.

The plane can be written in terms of the normal $\hat{n}$ as $x \cdot \hat{n} = \langle x, \hat{n} \rangle = 0$, with

$$\hat{n} = \frac{1}{3} \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}.$$

Then

$$R(v) = v - 2 \langle v, \hat{n} \rangle \hat{n} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} - 2 \left[\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \cdot \frac{1}{3} \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix} \right] \frac{1}{3} \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} -1 \\ 2 \\ 11 \end{pmatrix}.$$

Example Find the reflection of $(1, 0, 2, 1)$ in $U = \{x \mid x_1 + x_2 = 0, x_3 + x_4 = 0\}$.

We see that U is spanned by

$$u_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \\ 0 \\ 0 \end{pmatrix}, \quad u_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 0 \\ 1 \\ -1 \end{pmatrix}.$$

We can write

$$v = u + \tilde{u} \Rightarrow \tilde{u} = v - u = v - \langle v, u_i \rangle u_i,$$

then since

$$\langle v, u_1 \rangle = \frac{1}{\sqrt{2}}(v_1 - v_2), \quad \langle v, u_2 \rangle = \frac{1}{\sqrt{2}}(v_3 - v_4),$$

which provides $\tilde{u}$, from which

$$u = \frac{1}{2} \begin{pmatrix} v_1 - v_2 \\ v_2 - v_1 \\ v_3 - v_4 \\ v_4 - v_3 \end{pmatrix}.$$

Then

$$R(v) = u - \tilde{u} = 2u - v = - \begin{pmatrix} v_2 \\ v_1 \\ v_4 \\ v_3 \end{pmatrix} \Rightarrow R \left(\begin{pmatrix} 1 \\ 0 \\ 2 \\ 1 \end{pmatrix} \right) = \begin{pmatrix} 0 \\ -1 \\ -1 \\ -2 \end{pmatrix}.$$

Example Let $V = C([-1, 1])$ with the usual inner product. Let $U = \{f(t) \mid f(-t) = f(t) \forall t \in [-1, 1]\}$, i.e. the set of even functions on $[-1, 1]$. Then the $U^\perp$ is the set of odd functions on $[-1, 1]$. Then since

$$f(t) = \frac{1}{2}(f(t) + f(-t)) + \frac{1}{2}(f(t) - f(-t)) = f_e(t) + f_o(t)$$

where the bracketed terms are respectively even and odd. The reflection is thus

$$f(t) = f_e(t) - f_o(t) = f(-t),$$

which is what might have been expected.

If $\dim(V) = n$ and $\dim(U) = k$ with U a subspace of V , then we can find a basis $\{u_1, \dots, u_k, v_{k+1}, \dots, v_n\}$ of V so that

$$\begin{aligned} R(\lambda_1 u_1 + \dots \lambda_k u_k + \lambda_{k+1} v_{k+1} + \dots \lambda_n v_n) \\ = \lambda_1 u_1 + \dots \lambda_k u_k - \lambda_{k+1} v_{k+1} - \dots \lambda_n v_n. \end{aligned}$$

Then R is represented by

$$A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

where $\text{rank}(A) = n$ and $\text{signature}(A) = 2k - n$. Since A is symmetric and orthogonal, we see reflections are symmetric linear maps and an isometry. Note that $A^2 = I$ by the Cayley-Hamilton theorem since the eigenvalues of A are ± 1 .

In contrast, projections P are given by

$$\begin{aligned} P(\lambda_1 u_1 + \dots \lambda_k u_k + \lambda_{k+1} v_{k+1} + \dots \lambda_n v_n) \\ = \lambda_1 u_1 + \dots \lambda_k u_k, \end{aligned}$$

and the corresponding matrix B say has rank k and signature $2k - k = k$. Here $B^2 = B$ and $|B| = 0$, so they are symmetric linear maps but not more (and is clearly not invertible since projection “loses” information).

4.5 Orthogonal functions and polynomials

A linear map of an infinite dimensional space to itself is called an **operator**.

Consider the space $F[2\pi] = \{f \mid f : [-\pi, \pi] \rightarrow \mathbb{R}, f(-\pi) = f(\pi)\}$, i.e. the periodic functions over the interval $[-\pi, \pi]$, with the inner product

$$\langle f, g \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x)g(x) \, dx.$$

We want to investigate the differential operator

$$\frac{d^2}{dx^2} : F[2\pi] \rightarrow \mathbb{R}; f(x) \mapsto f''(x).$$

Proposition 4.5.1 d^2/dx^2 is a symmetric operator.

Proof Assuming the second derivatives exist, then we basically just do integration by parts twice, noting that the boundary terms vanish by periodicity. ■

This then implies all eigenvalues of the second derivative operator are real when acting on F .

Proposition 4.5.2 1. All eigenvalues of d^2/dx^2 are of the form $-n^2$, $n \in \mathbb{Z}$ (i.e. it is a negative semi-definite operator).

2. Any eigenfunction with eigenvalue $\lambda = 0$ is a constant.

3. All eigenfunctions with eigenvalue $\lambda \neq 0$ is a linear combination of $\sin n\pi x$ and $\cos n\pi x$

Usually we talk about $-d^2/dx^2$ so we talk about positive definite-ness.

Link with Fourier analysis coming up!

Proof First and third item can be shown together, with the second item as a special case. Suppose $\lambda_0 > 0$, then $\lambda_i = \mu_0^2$ where $\mu_0 \in \mathbb{R} \setminus \{0\}$, which implies the eigenfunctions are

$$y_0(x) = Ae^{\mu_0 x} + Be^{-\mu_0 x}.$$

Periodicity boundary conditions imply $A = B = 0$ so there are no solutions and $\lambda_0 < 0$. The same manipulations give

$$y_0(x) = Ce^{i\mu_0 x} + De^{-i\mu_0 x} = A \cos \mu_0 x + B \sin \mu_0 x,$$

and periodicity boundary conditions imply that $B = 0$ or $\mu_0 \in \mathbb{Z}$. The first one is incompatible, so eigenvalues are of the form $-\mu_0^2$ with $\mu_0 \in \mathbb{Z}$.

The special case $\lambda_0 = 0$ gives the eigenfunction $y_0(x) = Ax + B$, but periodicity implies $A = 0$, so we have constant functions. ■

Corollary 4.5.3 For integers n and m , $\cos nx$ (to include the non-zero constants being the special case of $n = 0$) and $\sin nx$ (to include zero for $m = 0$) in F are mutually orthogonal in the inner product above.

Proof For $m \neq n$ they above are eigenfunctions of a symmetric operator corresponding to different eigenvalues. For $m = n \neq 0$, the resulting function is an odd function over the interval. ■

Let $F_\mu([a, b])$ be the space of continuous functions from $[a, b]$ to $\mathbb{R}$ with a *weighted* inner product

$$\langle f, g \rangle = \int_a^b f(x)g(x)\mu(x) dx, \quad (4.5)$$

with $\mu(x) > 0$ for all $x \in (a, b)$.

Consider the space of all polynomial from $[a, b] \rightarrow \mathbb{R}$. Then $\{1, x, \dots, x^n, \dots\}$ is a basis of F_μ . Applying Gram-Schmidt on this set we can find an orthogonal system of polynomials $\{P_0(x), \dots, P_n(x), \dots\}$ with $\deg(P_n(x)) = n$. For any $f \in F_\mu([a, b])$ we can write $f(x) = \sum c_k P_k(x)$ where $c_i = \langle f, P_i \rangle / \|P_i\|$ for any $i \geq 0$. We can then see that

$$f_n(x) = \sum_{k=1}^n c_k P_k(x)$$

If additionally we can show the eigenfunctions form a basis then it basically says all periodic functions can be expanded as sines and cosines, which is the basis of Fourier analysis. This requires a little more subtlety since we are dealing with infinite dimension spaces and will not be done here (we need to restrict to L^2 functions).

forms the ‘best’ approximation of $f(x)$ with respect to the defined inner product.

The first commonly used systems are summarised in the table below.

polynomial	interval	weight $\mu(x)$
Legendre	$(-1, 1)$	1
Chebyshev-I	$(-1, 1)$	$1/\sqrt{1-x^2}$
Chebyshev-II	$(-1, 1)$	$\sqrt{1-x^2}$
Hermite	$(-\infty, \infty)$	e^{-x^2}
Laguerre	$(0, \infty)$	e^{-x}

These show up in numerical analysis and quantum mechanics quite a bit. These show up in **Sturm-Liouville problems** relating to self-adjoint operators.

Suppose $\mathcal{L}$ denotes a second order linear differential operator so that for $u \in F[(a, b)]$,

$$\mathcal{L}u = p_0 u'' + p_1 u' + p_2 u,$$

where p_i are real polynomial coefficients and the problem is non-singular (i.e. $p_0(x)$ is non-vanishing on (a, b)). We aim to find a weight function $\mu(x)$ so that $\mathcal{L}$ is symmetric with respect to $\langle \cdot, \cdot \rangle_\mu$. Let $\mu(x)$ be such that, for all $x \in (a, b)$,

$$\mu(x)p_1(x) = [\mu(x)p_0(x)]' = \mu(x)p_0'(x) + \mu'(x)p_0(x),$$

i.e. $\mu(x)$ the the integrating factor of the linear differential equation

$$\mu(x) = \exp \left[\int \frac{p_1 - p_0'}{p_0} d\tilde{x} \right] > 0,$$

which is also well-defined because p_0 is non-vanishing on the interval.

Theorem 4.5.4 *If $\mu(x)p_0(x)$ vanishes at $x = a$ and $x = b$, then $\mathcal{L}$ is symmetric with respect to $\langle \cdot, \cdot \rangle_\mu$.*

Again, boundary conditions are central to the problem...

Proof Let $p(x) = \mu(x)p_0(x)$, then $p(a) = p(b) = 0$ by assumption. Then

$$\mu(x)p_1(x) = [\mu(x)p_0(x)]' = p'(x),$$

which implies

$$\mu \mathcal{L}u = p(x)u''(x) + p'(x)u'(x) + \mu(x)p_2(x)u(x) = [pu']' + \mu p_2 u.$$

Contracting with another function $v \in F$ and integrating over the domain gives

$$\begin{aligned} \langle \mathcal{L}u, v \rangle_\mu &= \int_a^b (v[pu']' + v\mu p_2 u) dx \\ &= \int_a^b (-v'pu' + v\mu p_2 u) dx \end{aligned}$$

since the boundary terms vanish. The above expression is symmetric in u and v (or could do integration by parts one more time), demonstrating that $\langle \mathcal{L}u, v \rangle_\mu = \langle u, \mathcal{L}v \rangle_\mu$. ■

Theorem 4.5.5 Let $P_0(x) = 1, P_1(x), \dots, P_n(x), \dots$ be the non-normalised system of polynomials orthogonal to each other with respect to $\mathcal{L}$. If $u(x) \in \mathbb{R}[x]$, and suppose $\mathcal{L}u$ is a polynomial of degree at most $\deg(u)$, then the polynomials $P_0, P_1, \dots$ above are eigenfunctions of $\mathcal{L}$.

Proof Note that $\{P_0, \dots, P_n\}$ is a basis of $\mathbb{R}[x]$. By hypothesis, $\mathcal{L}P_n = c_n P_n$ (no sum implied) where P_n is an eigenfunction, so for some i ,

$$c_i \langle P_i, P_k \rangle = \langle \mathcal{L}P_n, P_k \rangle = \langle P_n, \mathcal{L}P_k \rangle = 0$$

if $k < n$. Since $\mathcal{L}P_k \in \mathbb{R}[x]$ and is a linear combination of $P_0, \dots, P_k$, and P_n is orthogonal to each of these, then $c_i = 0$, and since n was arbitrary, $P_0, P_1, \dots$ are eigenfunctions of $\mathcal{L}$. ■

Special cases of these are as follows:

case	interval	operator $\mathcal{L}$
Legendre	$(-1, 1)$	$(1 - x^2) \frac{d^2}{dx^2} - 2x \frac{d}{dx}$
Chebyshev-I	$(-1, 1)$	$(1 - x^2) \frac{d^2}{dx^2} - x \frac{d}{dx}$
Chebyshev-II	$(-1, 1)$	$(1 - x^2) \frac{d^2}{dx^2} - 3x \frac{d}{dx}$
Hermite	$(-\infty, \infty)$	$x \frac{d^2}{dx^2} - 2x \frac{d}{dx}$
Laguerre	$(0, \infty)$	$x \frac{d^2}{dx^2} + (1 - x) \frac{d}{dx}$

4.6 Jordan normal form

Let V be a vector space and $T : V \rightarrow V$ be a linear map. Given a basis of V , we can write T as a matrix with respect to such a basis. We get a different matrix depending on choice of matrix, however.

Example If $\dim(V) = n$ and T has n linearly independent eigenvectors, then with respect to the eigenvector basis $A = \text{diag}(\lambda_1, \dots, \lambda_n)$ represents T .

Let W be a subspace of V . If $T(W) \subset W$, then W is called an **invariant subspace** of T .

If V is a direct sum of subspaces, i.e. $V = W_1 \oplus W_2$ and $W_{1,2}$ are invariant subspaces of T , then T may be represented by a **block diagonal matrix**

$$A = \begin{pmatrix} A_1 & 0 \\ 0 & A_2 \end{pmatrix}$$

if we use a basis for V made up of a union of basis vectors of W_1 and W_2 .

Example If U is a subspace of V then $V = U \oplus U^\perp$, and a reflection T can be represented as (in a suitable basis)

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

If $V = W_1 \oplus \dots \oplus W_k$ under some basis of V , then we have the block diagonal matrix

$$A = \begin{pmatrix} A_1 & & & 0 \\ & A_2 & & \\ & & \ddots & \\ 0 & & & A_k \end{pmatrix}.$$

Suppose the characteristic polynomial of T is

$$\prod_{i=1}^k (t - \lambda_i)^{s_i}, \quad s_i \in \mathbb{Z}, \lambda_i \in \mathbb{C}.$$

We can write $V = W_1 \oplus \dots \oplus W_k$ where each W_j is given by $W_j = \{v \in V \mid (A - \lambda_j I)^{s_j}(v) = \mathbf{0}\}$. If $s_j = 1$ then this is equivalent to saying W_k is the eigenspace spanned by eigenvector with eigenvalue λ_j .

Example If $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is represented by

$$A = \begin{pmatrix} 3 & -2 \\ 2 & -1 \end{pmatrix}$$

with respect to the standard basis, then $|tI - A| = (t - 1)^2$, and the eigenvector is of the form

$$x = \begin{pmatrix} x \\ x \end{pmatrix}, \quad x \in \mathbb{R}.$$

We also see that $(I - A)^2 = O$, so we can find another vector solving the equation, and together with $(1, 1)$ forms a basis. We see

$$Ax_1 = x_1, \quad Ax_2 = a_1x_1 + a_2x_2,$$

and

$$\mathbf{0} = (A - I)^2x_2 = (A - I)[(A - I)x_2] - (A - I)y,$$

so

$$y = tx_1 = (A - I)x_2 = Ax_2 - x_2$$

implying

$$tx_1 + x_2 = Ax_2.$$

It is always possible to scale, giving $t = 1$, so with respect to x_1 and x_2 we have

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Theorem 4.6.1 Let $T : V \rightarrow V$ be a linear map with characteristic polynomial

$$\prod_{i=1}^k (t - \lambda_i)^{s_i}, \quad s_i \in \mathbb{Z}, \lambda_i \in \mathbb{C}.$$

Then there is a basis of V for which T is given by the block diagonal matrix

$$A = \begin{pmatrix} A_1 & & 0 \\ & A_2 & \\ & & \ddots \\ 0 & & & A_k \end{pmatrix},$$

where each A_i corresponds to eigenvalue λ_j , and they are all of **Jordan block form**, i.e.

$$(\lambda_j), \quad \begin{pmatrix} \lambda_j & 1 \\ 0 & \lambda_j \end{pmatrix}, \quad \begin{pmatrix} \lambda_j & 1 & 0 \\ 0 & \lambda_j & 1 \\ 0 & 0 & \lambda_j \end{pmatrix}, \dots$$

□

Example Find the Jordan normal form of

$$\begin{pmatrix} 2 & 2 & -2 \\ 1 & -1 & 1 \\ 2 & 0 & 0 \end{pmatrix}.$$

The characteristic polynomial is $\lambda^2(\lambda - 1) = 0$, so there is a degenerate case. The two eigenvectors are

$$u_1 = \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}, \quad u_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix},$$

but we need a third vector u_3 for the 0-eigenspace with $Au_3 = 0u_3 + u_2 = u_2$, which implies $A^2u_3 = 0 = x_1 + x_2 - x_3$. We see that gives

$$x_3 = \begin{pmatrix} x_1 \\ x_2 \\ x_1 + x_2 \end{pmatrix},$$

and

$$\mathbf{u}_2 = \mathbf{A}\mathbf{x} = \begin{pmatrix} 0 \\ 2x_1 \\ 2x_1 \end{pmatrix},$$

so $x_1 = 1/2$, and letting $x_2 = 1/2$ gives

$$\mathbf{u}_3 = \begin{pmatrix} 1/2 \\ -1/2 \\ 0 \end{pmatrix}.$$

This gives

$$\mathbf{P} = \begin{pmatrix} 2 & 0 & 1/2 \\ 3 & 1 & -1/2 \\ 4 & 1 & 0 \end{pmatrix}, \quad \mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$